

Selecting for Restraint:

Personalist Power and Delegated Political Selection

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How does personalist power travel downward through rulers' trusted agents? I argue that personalist rule changes not only whom rulers appoint, but also how those agents select subordinates. Because protégés depend on the ruler's ongoing trust, subordinate achievements generate limited independent political capital, while failures travel upward, damaging their standing as trustworthy agents. Delegated selection turns on a third criterion beyond loyalty and competence: the political risk of how subordinates produce performance, leading protégés to favor controllable and defensible governing styles. I test this at China's province–prefecture interface under Xi Jinping. City leaders promoted by Xi-protégé provincial party secretaries show lower reported GDP growth relative to external benchmarks, but no comparable gap in less manipulable indicators. Earlier mayoral records further reveal restrained fiscal and land-based trajectories before selection. The findings point to a selection logic of restraint, not lower capacity, through which personalist power is reproduced below the ruler.

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Introduction

Personalist rule has become central to contemporary authoritarianism. Since the end of the Cold War, personalist dictatorships have grown from a minority of autocracies into one of authoritarianism's dominant forms (Geddes, Wright, and Frantz 2014; Kendall-Taylor, Frantz, and Wright 2017). This rise creates a central organizational problem for authoritarian governance. By weakening elite constraints, centralizing appointment authority, sidelining rivals, and empowering loyalists, dictators can make the hierarchy more responsive to their commands (Baturu and Elkink 2016; Gandhi and Sumner 2020; Geddes, Wright, and Frantz 2018; Svolik 2009, 2012; Timoneda, Escribà-Folch, and Haynes 2025). Yet the same concentration of authority that strengthens command can also undermine governance. Personalist rule can narrow information flows and discourage autonomous local initiative, and it has been associated with lower economic performance and poorer public-goods provision (Blattman, Gehlbach, and Yu 2025; Egorov and Sonin 2011; Geddes, Wright, and Frantz 2018; Wintrobe 1998). Why might a hierarchy become more responsive to the ruler yet less capable of sustaining developmental dynamism?

Existing explanations often emphasize two related mechanisms. One emphasizes institutional weakening and subordination, as personalist rulers subordinate the institutions that govern elite appointment, promotion, and punishment to more discretionary personal control (Gandhi and Sumner 2020; Geddes, Wright, and Frantz 2018; Svolik 2012; Timoneda, Escribà-Folch, and Haynes 2025). A second emphasizes the loyalty–competence trade-off, since insecure rulers may sacrifice administrative competence in favor of officials who are loyal, politically dependent, or less threatening (Egorov and Sonin 2011; Wintrobe 1998; Zakharov 2016).

However, no ruler governs through the inner circle alone. Even highly centralized dictators must delegate lower-level personnel control to trusted agents with local information and appointment authority. Once loyalists and protégés occupy pivotal personnel posts, they become selectors in their own right. This delegation matters especially at the subnational level, where officials maintain political stability, implement policy, gather information, manage local economies, and allocate

public resources (Landry 2008; Reuter and Robertson 2012; Reuter et al. 2016). The organizational consequences of personalism therefore depend not only on whether trusted agents obey the ruler, but also on how they exercise personnel authority below the top leadership.

I argue that delegated selection under personalism turns on a third criterion beyond loyalty and competence: the political risk profile of how subordinates produce performance. Subordinates differ not only in loyalty or competence, but also in whether their achievements are generated through strategies that are controllable, defensible, and unlikely to expose superiors to blame. Because protégés' standing depends more on preserving the ruler's trust than on accumulating independent reputational capital, subordinate achievements offer limited durable credit, while subordinate failures can travel upward as evidence of poor judgment or weak oversight, undermining their standing as trustworthy agents. They therefore favor restraint, a politically defensible governing style that avoids high-variance, hard-to-verify strategies that create fiscal, statistical, or political exposure. Restrained officials can implement priorities and maintain order, but steer clear of risky local initiative. Such selection may narrow which governing styles reach office, dampening local developmental dynamism even when the officials chosen are no less capable.

Empirically, I test this argument at China's province–prefecture interface under Xi Jinping's personalization of power, examining how Xi-protégé provincial party secretaries select prefecture-level city leaders. I find that their appointees have lower leader-level performance profiles when performance is measured by reported GDP growth, but not when it is measured by less directly manipulable indicators such as loan growth or nighttime-light growth. Comparable appointments under provincial party secretaries connected to Xi's predecessor, Hu Jintao, do not show the same pattern, suggesting that the Xi-era result is not simply a generic effect of connection to the incumbent top leader. A GDP–nighttime-light residual profile further clarifies the interpretation. Xi-protégé appointees are associated with lower reported GDP growth relative to the luminosity benchmark, consistent with greater statistical restraint rather than a broad deficit in underlying economic capacity.

To assess whether this pattern reflects governing records observed before later selection rather

than post-selection adaptation, and whether restraint extends beyond statistical discipline, I examine the earlier mayoral spells of officials who later became first-time prefectural party secretaries. These pre-selection analyses show that officials later promoted under Xi-protégé provincial party secretaries had already followed more restrained fiscal and land-based development trajectories, with slower fiscal-expenditure growth and a lower likelihood of land-revenue expansion.

The central contribution of this article is to recast personalization as a hierarchical process of organizational reproduction rather than only a top-level problem of elite control. Existing research explains how rulers personalize power by weakening constraints, centralizing appointment authority, and elevating loyalists (Gandhi and Sumner 2020; Geddes, Wright, and Frantz 2018; Svolik 2012; Timoneda, Escribà-Folch, and Haynes 2025). I show that the consequences of personalization do not stop once loyalists are appointed. When trusted agents acquire personnel authority, they become delegated selectors, and their own dependence on the ruler shapes the criteria they use to staff lower levels of the state. Personalist power may therefore travel not only through the ruler's direct appointments, but also through the downstream selection criteria used by politically dependent agents. This perspective shifts attention from the ruler's consolidation of power to the organizational channels through which personalist authority travels below the top leadership.

This perspective also extends accounts of the “personalist penalty” (Blattman, Gehlbach, and Yu 2025). Existing explanations often link personalist rule to poorer governance and weaker economic outcomes through the elevation of loyal but less capable agents, weaker public-goods provision, lower private investment, greater conflict, or poorer decision-making (Besley and Kudamatsu 2007; Bueno de Mesquita et al. 2003; Egorov and Sonin 2026; Gandhi 2008; Gehlbach and Keefer 2011, 2012; Jackson and Morelli 2007; Svolik 2012; Weeks 2012; Wintrobe 1998; Wright 2008). The mechanism identified here operates at the level of political selection. Personalist rule may reshape governance not only by elevating loyalists with lower administrative capacity, but also by changing which forms of competence and initiative are politically acceptable. Officials associated with high-variance, politically exposed strategies of performance production may be discounted, while more cautious and defensible governing styles are favored. This selection logic can narrow

the range of acceptable bureaucratic entrepreneurship without requiring the selection of systematically less capable officials.

Finally, the article extends research on political selection in reform-era China. Existing studies have largely examined whether cadre advancement rewards performance under formal personnel institutions or patronage through informal personal connections (Jia, Kudamatsu, and Seim 2015; Keller 2016; Landry 2008; Landry, Lü, and Duan 2018; Li and Zhou 2005; Manion 2023; Shih, Adolph, and Liu 2012; Xu 2011; Yao and Zhang 2015). Recent work further shows that selection can reflect tacit, appearance-based impressions of candidates (Jiang and Yang 2026). I bring this debate into China's personalist turn by showing that appointments under politically empowered protégés reveal a distinct selection logic in which restraint becomes a politically valuable selection criterion beyond the conventional contrast between connections and performance.

Theory

Personalization of Power and Delegated Personnel Control

Authoritarian rule depends on how power is distributed within the ruling coalition. In some regimes, power remains relatively balanced between the ruler and other ruling elites, who retain organizational resources, coercive access, factional networks, or institutional leverage that can constrain or remove him. In others, the ruler has marginalized these elites and concentrated authority to the point that they can no longer credibly threaten his rule (Gandhi and Sumner 2020; Svoblik 2012). Personalization of power is the process through which rulers move toward the latter arrangement by weakening autonomous elite constraints and embedding political authority more directly in their own discretion.

This process grows out of a fundamental problem of authoritarian power sharing. Rulers need empowered insiders to govern, but those same insiders can use administrative resources, coercive organizations, and personnel authority to build independent bases of power (Paine 2021; Svoblik 2009; Tullock 1987). Personalization addresses this problem by shifting authority away from relatively autonomous elites and toward agents whose positions depend more directly on the ruler's

confidence. Through loyalist appointments, elite turnover, and institutional counterweights, rulers reduce the capacity of powerful insiders to entrench independent bases while expanding the role of politically dependent agents in the regime hierarchy (Baturu and Elkink 2016; Bokobza et al. 2022; Geddes, Wright, and Frantz 2018; Goldring and Matthews 2023; Mahdavi and Ishiyama 2020; Sudduth 2017).

This solution, however, creates a further problem of delegated governance. Even highly centralized dictators cannot personally observe, evaluate, and select all lower-level officials; they must rely on trusted agents who hold local information and formal authority over appointments, evaluations, and promotions (Jiang 2018; Landry 2008; Svolik 2012). As autonomous elites are marginalized, politically dependent loyalists and protégés are elevated to pivotal offices where they make personnel decisions of their own. The consequences of personalization therefore depend not only on whom the ruler appoints, but also on how these agents exercise personnel authority below the top leadership. This creates a second-order problem of political selection: do empowered loyalists reproduce the ruler's loyalty-first logic, prioritize officials who generate visible performance, or adopt a distinct selection logic shaped by their own position within the personalist hierarchy?

I argue that delegated selection under personalism is shaped by the political risk profile of how subordinates produce performance. Because protégés hold authority that is derivative of the ruler's trust and exposed to blame from below, they evaluate subordinates not only by loyalty or competence, but also by whether their governing strategies are controllable, defensible, and unlikely to expose superiors to blame. The result is a selection logic of restraint.

Derivative Authority and Blame Exposure under Personalist Delegation

Protégés occupy a distinctive position within personalist rule because their authority is derivative. They do not hold office primarily through independent constituencies, autonomous organizational bases, or reputational capital that can survive the withdrawal of the ruler's favor. Their standing rests on continued personal trust, access, and perceived usefulness within a ruler-centered hierarchy, a dependence reflected in elite incentives to profess loyalty under autocratic rule (Baturu, Khokhlov, and Tolstrup 2025). This does not require assuming that protégés subjectively feel insecure in every

moment. Vulnerability is not a psychological state here, but a structural position in which protégés hold delegated authority that can be revoked from above while remaining responsible for failures generated below.

This dependence is temporal as well as organizational. Because protégés owe their rise to the ruler's patronage, their fortunes are tied to the ruler's political survival. They are less likely to remain valuable to a successor once their patron loses power (Meng 2020; Oliveros 2021). Their dependence deepens as personalization advances, because the ruler's expanding control over appointments, resources, and protection leaves even favored protégés reliant on discretionary trust (Baturu and Elkind 2016). Protégés are therefore both beneficiaries and instruments of personalization: they gain authority from the ruler, but their continued access to office depends on exercising that authority in politically reliable ways (S. Chan 2003; Geddes, Wright, and Frantz 2018).

At first glance, the incentives seem to pull the other way. Because protégés must prove their usefulness to the ruler, one might expect them to favor aggressive, high-output subordinates who deliver visible achievements. Personalist delegation, however, changes both sides of the calculation. On the upside, the value of subordinate success depends on the political currency that sustains the selector's own position. For selectors with more autonomous standing, successful appointees can feed reputational capital, factional standing, and claims to governing capacity. For protégés, by contrast, career security and future advancement depend more directly on preserving the ruler's trust. Subordinate achievements can still signal usefulness, but they carry lower marginal value for the political currency on which protégés principally depend. Nor can protégés safely convert such success into independent standing: in a ruler-centered hierarchy, accumulating an autonomous base can itself invite suspicion, so subordinate achievements remain conditional signals of usefulness rather than durable political assets (Egorov and Sonin 2011; Sudduth 2017).

On the downside, a subordinate who produces visible achievements through aggressive policy entrepreneurship may also generate inflated statistics, fiscal stress, corruption exposure, social instability, or policy controversy. Once such problems surface, they can travel upward as evidence that the superior selected poorly, monitored weakly, or tolerated politically unsafe governance. More

autonomous selectors face a different calculus because subordinate achievements can more easily build reputation or factional standing, while independent bases can cushion blame from failure. For protégés, by contrast, the downside risks of a failed appointment are harder to cushion than the credit from subordinate success is to convert into durable political capital. The contrast makes high-variance, high-output subordinates politically riskier for protégés than for more autonomous selectors, tilting delegated selection toward blame avoidance rather than the pursuit of visible wins alone (Hood 2011; Weaver 1986).

This logic differs from accounts in which informal patronage ties strengthen clients' performance incentives through credible rewards, extra-institutional protection, and pressure to serve patrons' policy goals (Jiang 2018). One might expect the opposite: because protégés enjoy the ruler's favor, they should be especially well positioned to shield subordinates, absorb appointment failures, and tolerate risk. But this protection logic treats the source of a protégé's authority as if it were an unconditional guarantee. Here, the relevant selector is himself a dependent agent. His capacity to protect subordinates remains contingent on the ruler's continued trust, and subordinate failures are precisely the events that can weaken that trust.

The asymmetry turns on political currency as much as magnitude. Consider a protégé whose appointee delivers visible gains through aggressive policy entrepreneurship. The achievement may earn the selector marginal credit as a judge of talent, but that credit accrues to competence, not to the trust on which his standing rests; a more autonomous selector could convert the same success into reputation or factional standing. Failure cuts differently. If the same appointee's aggressive strategy instead triggers social unrest, the ruler can read the failure as poor judgment, as weak discipline over appointees, or as evidence that the protégé may be using his appointments to cultivate a power base of his own, a charge that bears on the trust relationship itself rather than on ordinary competence.

Personalist rule further sharpens this asymmetry. Because criticism of the ruler's own directives is politically constrained, policy failures can be recast as failures of implementation, judgment, or discipline by the ruler's agents. Delegated selectors and their appointees therefore become available sites for blame displacement. Rotation, reshuffling, demotion, and purge reinforce this logic

by reminding protégés that their authority remains conditional on continued political usefulness (Mahdavi and Ishiyama 2020; Sudduth 2017). Protégés thus face a problem of controlled delegation. They need subordinates who can implement policy and produce outcomes, but whose governing strategies remain monitorable, defensible, and unlikely to expose the selector to blame.

Subordinate Performance as Political Risk

This argument moves beyond the conventional loyalty–competence trade-off. Existing accounts often treat personalist selection as a choice between capable but potentially threatening officials and loyal but less capable ones (Egorov and Sonin 2011; Wintrobe 1998; Zakharov 2016). That trade-off is important, but incomplete. Delegated selection under personalism introduces a third criterion, the political risk profile of how subordinates produce performance. Subordinates differ not only in how loyal or competent they are, but also in the strategies through which they generate observable achievements. Some forms of performance production are routine, auditable, and politically defensible. Others are high-variance, hard to audit in real time, and likely to generate upward blame if they later fail.

I call the governing style favored by this selection logic restraint. Restraint is distinct from both loyalty and competence. Loyalty concerns political alignment with a superior, competence concerns underlying capacity, and restraint concerns the mode through which performance is produced under hierarchical delegation. Two officials may be similarly capable and politically reliable, yet differ sharply in restraint if one generates achievements through controllable and defensible strategies while the other relies on aggressive, difficult-to-monitor, or hard-to-contain forms of initiative. A loyal subordinate can be unrestrained if he pursues risky strategies to demonstrate zeal, please superiors, or stand out in local competition. A competent subordinate can also be unrestrained if his competence lies in producing visible achievements through methods that create hidden liabilities for superiors. Conversely, a restrained official need not be weak or inactive. He may implement priorities and maintain order, but do so through strategies that are easier for superiors to monitor, defend, and contain.

This distinction matters especially in authoritarian bureaucracies, where performance signals

are politically valuable but imperfectly observable (Lorentzen 2013; Wallace 2016). The problem is not simply that superiors know nothing about how performance is produced. Rather, the risks embedded in performance production may be difficult to audit in real time and costly to defend once problems surface. Achievements generated through aggressive or hard-to-monitor strategies may look impressive in the short run, but they can later expose the selector to blame if investigations, policy reversals, social conflicts, or fiscal problems reveal that the subordinate was politically unsafe. A subordinate who pursues such strategies may also accumulate local autonomy or an independent power base, itself a source of risk for the superior.

This does not imply that protégés favor passive subordinates or avoid policy effort. On the contrary, protégés may be especially eager to see the ruler’s priorities implemented, because their own careers depend on demonstrating responsiveness to the center. More generally, patronage ties within centralized hierarchies can redirect subordinate effort toward politically salient tasks, even when doing so creates trade-offs in other governance domains (Feng et al. 2025). Yet compliance-oriented activism differs from local developmental entrepreneurship. In domains that require discretion, experimentation, and local initiative, protégés have stronger incentives to favor officials whose conduct is controllable and defensible. They may still value competence, but they should discount candidates whose records suggest ambitious, autonomy-seeking, or politically exposed forms of performance production.

This logic points to a distinctive empirical pattern. If protégés select for restraint rather than low capacity, their appointees should not appear uniformly weaker across observable indicators of governance performance. Selection gaps should be concentrated in domains where officials have greater discretion over how performance is produced, where achievements are difficult to verify fully ex ante, and where later revelations of manipulation or hidden liabilities are costly for superiors to defend. Where indicators are more independently recorded and less directly subject to managerial discretion, the gap should weaken or disappear.

Empirical Setting: Personalist Power and Subnational Political Selection in China

Xi's Personalization of Power and the Rise of Protégé Provincial Leaders

I test the argument of “selecting for restraint” in the context of Xi Jinping’s personalization of power in China. Xi succeeded Hu Jintao as General Secretary of the Chinese Communist Party (CCP) and Chairman of the Central Military Commission in November 2012, and assumed the presidency of the People’s Republic of China (PRC) in March 2013. At first, this transition appeared to extend the post-Mao pattern of institutionalized succession and collective leadership. Yet Xi’s first decade in power marked a sharp reversal of that trajectory. Through a sweeping anticorruption and disciplinary campaign, tighter ideological control, the restructuring of the People’s Liberation Army, and direct leadership over key leading small groups and commissions, Xi centralized authority across the party-state. The abolition of presidential term limits in 2018, followed by Xi’s third term as CCP General Secretary in 2022 and PRC President in 2023, further signaled the erosion of post-Mao succession constraints and the return of more personalistic rule (Brødsgaard [2018](#); Fewsmith and Nathan [2019](#); L. Li [2019](#); J. Li [2022](#); Shirk [2018](#)).

For this study, the crucial feature of Xi’s personalization is not only that authority became more centralized at the top, but also that personnel control became increasingly organized around the ruler’s trusted agents. Existing studies show that Xi-era elite politics shifted from factional balance toward a dominant faction, that Xi promoted his clients more successfully than Jiang Zemin or Hu Jintao, and that personal connections became especially important in the appointment and promotion of provincial top leaders, including provincial party secretaries and governors (Choi, Givens, and MacDonald [2021](#); Lee [2025](#)). These changes make Xi-era China a useful setting for examining whether the personalist turn altered the logic of subnational political selection through the appointment choices of Xi’s protégés.

Provincial Party Secretaries and Subnational Personnel Authority

China's cadre management system makes this process observable at the province–prefecture interface. Since the early reform era, the CCP has operated a nomenklatura-based “one-level-down” personnel system, under which Party authorities at each administrative level manage key leadership positions one level below them (H. S. Chan 2004; Landry 2008; Manion 1985; Zeng 2016). At the provincial level, this authority covers a broad set of prefecture- and bureau-level positions, including provincial department heads and key leadership posts in prefecture-level cities. Among these positions, I focus on prefecture-level city party secretaries and mayors, the two leading executives responsible for local political control and economic governance. Provincial party secretaries are especially important in this process because they sit at the apex of provincial personnel politics and typically exercise the greatest influence over prefecture-level appointments (Jiang 2018; Jiang and Zhang 2020; Landry 2008; Zeng 2016). Prefecture-level city leaders therefore provide a suitable site for the empirical analysis: their careers and governing records shape local development, but their appointments are normally mediated by provincial authorities rather than directly controlled by the national leadership (Landry 2008; Yao and Zhang 2015).

I use this institutional structure to compare prefecture-level city leaders promoted under Xi-era provincial party secretaries who are Xi's protégés with those promoted under provincial party secretaries who are not. The comparison captures a key mechanism of the personalist turn: as the ruler places trusted agents in pivotal subnational personnel positions, their appointment choices reveal how personalist power is transmitted downward into the bureaucracy.

Identifying Xi's Protégé Provincial Party Secretaries

To identify Xi's protégé provincial party secretaries, I use a conservative two-step coding strategy that combines prior opportunities for patron-client tie formation with subsequent elevation to provincial party secretary under Xi. Patron-client ties in elite politics are not directly observable, so the first step is not intended to measure private closeness to Xi Jinping. Instead, it identifies observable conditions under which hierarchical patron-client ties were plausible. Following Scott's

(1972) classic account, patron-client relations are characterized by inequality, face-to-face interaction, and diffuse flexibility. I therefore begin by identifying officials who had overlapping work experience with Xi in the same locality or administrative unit and who stood in a clear hierarchical relationship to him. Specifically, I apply a “one-level-down” principle. Potential protégés must have either served directly under Xi or held a position no more than one administrative rank below him before he became general secretary, during his tenure in Hebei, Fujian, Zhejiang, Shanghai, or as president of the Central Party School. This restriction is intended to capture relationships in which personal contact and hierarchical dependence were plausible, rather than merely loose regional, biographical, or factional affinity. It also makes the coding deliberately conservative: some genuine but less observable ties may be missed, but the rule reduces the risk of classifying officials as protégés on the basis of broad factional labels or routine bureaucratic proximity alone.

I then impose a second criterion. Among these officials, I code as Xi protégés only those who were later promoted to provincial party secretary after Xi became China’s top leader. This promotion-based criterion is important because prior overlap alone may capture weak acquaintance or routine bureaucratic contact (Jiang 2018). Subsequent elevation to provincial party secretary, one of the most important subnational posts and a position subject to top-level personnel authority, provides observable evidence that the earlier tie was politically consequential. The second criterion therefore distinguishes substantive political ties from mere prior contact and links the coding directly to the article’s focus on delegated personnel authority. Conversely, officials promoted to provincial party secretary under Xi are not automatically coded as protégés unless they also satisfy the prior work-overlap criterion, because career advancement alone may reflect competence, rotation, regional needs, or central personnel balancing. Applying these two criteria identifies 20 protégés who both had plausible opportunities to form patron-client ties with Xi and were later entrusted with provincial party secretary posts under his rule.

In addition to this work-experience-based rule, I include Li Xi as a literature-based case. Although Li Xi is not captured by the one-level-down work-overlap rule, existing accounts identify Shaanxi ties as an important component of Xi Jinping’s political network. These ties are politically

meaningful not simply because of birthplace, but because Shaanxi links Xi to both family-origin and formative-experience networks. Xi Zhongxun, Xi Jinping’s father, was born in Shaanxi and built much of his early revolutionary career there, while Xi Jinping himself spent his sent-down youth years in Yan’an (C. Li 2014a). Existing work therefore treats Shaanxi connections, formative-year ties, and overlapping work experience as major bases of Xi’s factional network (Choi, Givens, and MacDonald 2021; C. Li 2014b). Since Li Xi is identified as Shaanxi-linked and was elevated to provincial party secretary under Xi, I include him as a separately justified literature-based case rather than as part of the work-overlap rule.

In total, the coding identifies 21 Xi protégés, who together held 28 provincial party secretary tenures observed during the study period. Figure 1 shows the temporal and geographic expansion of their appointments across provincial party secretary posts from 2014 to 2024. Table A.1 in the Supplemental Materials lists all coded tenures held by Xi-protégé provincial party secretaries.

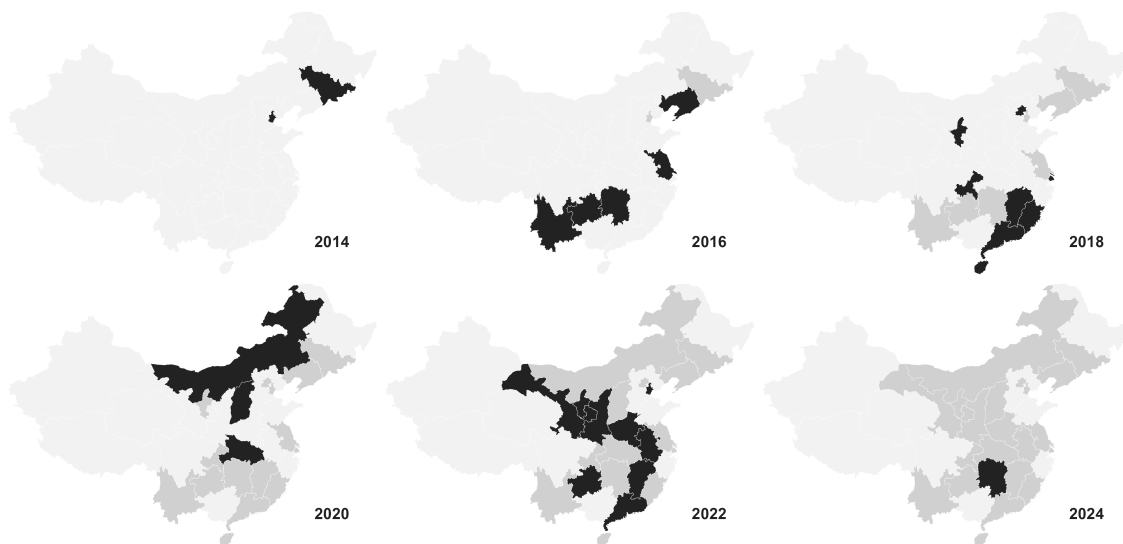


Figure 1: Xi-Protégé Provincial Party Secretaries, 2014–2024

Notes: Black marks provinces where Xi protégés were newly appointed as provincial party secretaries in each two-year interval, while light gray marks provinces previously held by Xi protégés. Panel labels indicate endpoint years.

Research Design

Growth Incentives and the Political Risk of GDP Inflation

Economic growth has long been politically consequential in China's cadre evaluation system, but its promotion payoff is neither uniform nor uncontested. At the highest levels of the party hierarchy, growth performance appears to matter less than factional ties, education, and fiscal revenue collection (Shih, Adolph, and Liu 2012). At the provincial level, earlier studies find evidence that economic performance affects leaders' promotion and termination prospects, including through relative performance evaluation, although subsequent work shows that these rewards are conditioned by political connections and other elite-level considerations (Chen, Li, and Zhou 2005; Jia, Kudamatsu, and Seim 2015; Li and Zhou 2005). At the city level and below, growth targets and performance metrics are more directly embedded in everyday cadre evaluation and local growth competition, making reported economic performance a visible and politically important signal even if promotion decisions are not mechanically determined by growth alone (Landry 2008; Landry, Lü, and Duan 2018; Yao and Zhang 2015).

Precisely because reported GDP mattered politically, local officials had incentives to inflate or strategically manage reported growth, especially when doing so helped them meet official growth targets (W. Chen et al. 2019; Lyu et al. 2018; Wallace 2016; Zeng and Zhou 2024). Yet GDP manipulation was not without political cost. Authoritarian central authorities require reliable local information to monitor subnational officials, assess policy implementation, and distinguish genuine performance from bureaucratic compliance theater (Dimitrov 2023; Lorentzen 2013; Malesky and Schuler 2011). Local GDP inflation undermines this informational function. It does not merely embellish local performance; it contaminates the statistics through which higher-level authorities evaluate cadres and diagnose local economic conditions. In this sense, local statistical manipulation creates a political problem for the center, even when favorable growth figures may appear useful for regime legitimation.

For provincial party secretaries, this problem was especially acute. As the officials responsible

for supervising prefectural leaders and shaping their promotion prospects, they could be blamed when inflated statistics distorted the local promotion environment. In Liaoning, after provincial authorities acknowledged inflated fiscal-revenue figures and problems in reported economic growth, official accounts associated the problem with the political environment under former provincial party secretary Wang Min, in which “numbers produced officials, and officials produced numbers.” The criticism implied that statistical falsification was not merely a technical reporting problem, but a failure of political supervision and cadre management.¹

Such blame was not necessarily arbitrary. Although provincial party secretaries did not observe local output perfectly, they were not simply passive recipients of prefectural GDP reports. Given their supervisory authority over prefectural officials and access to alternative administrative indicators, provincial leaders had channels through which they could assess the credibility of local performance claims and learn about officials’ broader governing styles. Li Keqiang’s remarks on Liaoning are illustrative. While serving as Liaoning’s provincial party secretary in 2007, Li, later China’s premier, reportedly described GDP figures as “man-made” and therefore unreliable, explaining that he relied instead on electricity consumption, rail cargo volume, and bank lending to assess economic conditions (Lyu et al. 2018; Wallace 2016).

This argument points to a clear observable implication in the Chinese context. Reported GDP growth is a useful domain for observing selection for restraint because it is politically salient, highly visible, and susceptible to exaggeration. Ambitious city leaders may rely on inflated growth figures to signal developmental capacity. Yet whether such signals attract or deter a selector depends on the credit–blame calculus developed above. Selectors with more autonomous political bases can more readily convert subordinate achievements into political capital and absorb later fallout, making such risks less politically prohibitive. For Xi’s protégés, by contrast, whose authority rests on the ruler’s

1. “辽宁省委：切实解决经济数据造假问题” [Liaoning Provincial Party Committee: Effectively Resolve the Problem of Economic Data Falsification], *The Paper*, August 27, 2016, accessed June 4, 2026, https://www.thepaper.cn/newsDetail_forward_1520142; “震惊 | 辽宁首次公开确认存在经济数据造假” [Shocking: Liaoning Publicly Confirms the Existence of Economic Data Falsification for the First Time], *CCTV News*, January 18, 2017, accessed June 4, 2026, <https://m.news.cctv.com/2017/01/18/ARTIAUDsHnWWb6wZjiLsY7zr170118.shtml>.

trust, the prospective blame from appointing officials with hard-to-verify performance-producing styles outweighs the durable credit they can credibly bank from those officials' achievements. If Xi's protégés select for restraint, then city leaders promoted under Xi-protégé provincial party secretaries should exhibit lower reported GDP performance, but this negative pattern should not be mirrored in less directly manipulable indicators of local economic activity.

The Hu-Era Benchmark

A central design concern is that this observable implication may capture connection to the incumbent top leader rather than the distinctive consequences of personalization. If provincial party secretaries connected to any sitting general secretary tend to promote more restrained city leaders, then the pattern observed under Xi's protégés would not necessarily reflect the personalist turn. To assess this possibility, I incorporate a Hu-era benchmark of provincial party secretaries connected to Hu Jintao. This benchmark provides leverage for assessing whether connected promoters under a more collective leadership arrangement produced the same selection pattern, or whether the pattern is specific to appointments made by Xi's protégés under personalized rule.

The Hu era is especially useful for this purpose because Hu Jintao had identifiable factional associates but governed under a more collective and power-sharing elite arrangement. I therefore refer to these officials as Hu-connected provincial party secretaries rather than Hu's protégés. Cheng Li's (2002) account of Hu's Youth League associates emphasizes that Hu was linked to officials with Chinese Communist Youth League (CCYL) backgrounds, but that these ties were based more on shared organizational identity and factional association than on personalist dependence. Similarly, Choi, Givens, and MacDonald (2021) characterize Hu's network as a Youth League or Hu-Li faction and contrast the factional balance of the Hu period with the emergence of a dominant Xi faction under Xi Jinping. The Hu-era comparison therefore provides a conservative benchmark: it tests whether incumbent connection, without the same degree of personalization, produces the same selection pattern.

I identify Hu-connected provincial party secretaries through an analogous two-step coding strategy. First, I identify officials with plausible prior ties to Hu Jintao. These ties include offi-

cials who served in a one-level-down relationship under Hu before he became general secretary, during his tenure in Gansu, Guizhou, Tibet, or as president of the Central Party School, as well as officials who held leadership posts in provincial-level CCYL committees or central-level CCYL organs before Hu became general secretary. This first step captures either direct hierarchical contact or membership in Hu's broader Youth League-based factional network. Second, among officials with prior ties to Hu, I code as Hu-connected provincial party secretaries only those who were elevated to provincial party secretary during Hu's leadership period. This promotion-based criterion parallels the Xi-era coding strategy and provides observable evidence that a prior factional tie became politically consequential.²

I attribute the provincial party secretary appointments made in March 2013 to Hu-era personnel arrangements. Although Xi became CCP general secretary in November 2012, these appointments occurred in March 2013, when Hu Jintao formally stepped down as state president and the Hu-to-Xi leadership transition was completed. I therefore code these March 2013 appointments as part of the Hu-era transition sequence, rather than treating them as ordinary post-transition Xi-era provincial appointments. Consistent with Jiang's (2018) promotion-based approach to patronage measurement, I assign appointments to the political leadership arrangement under which the relevant promotion decision was made, not simply to the calendar year in which the appointee began service. Applying these criteria identifies 19 Hu-connected provincial party secretaries, corresponding to 26 provincial party-secretary tenures initiated under Hu-era personnel arrangements between 2004 and 2013. Table A.2 in the Supplemental Materials lists all coded tenures held by Hu-connected provincial party secretaries.

The Hu-era benchmark sharpens the empirical test by assessing whether the observed pattern

2. Consistent with the organizational basis of Hu's network, most provincial party secretaries in the final Hu-connected benchmark sample are identified through the CCYL criterion rather than direct local work overlap. A strict one-level-down local work-overlap criterion would identify only two unique Hu-connected provincial party secretaries after applying the promotion-based criterion: Wang Sanyun through prior overlap with Hu in Guizhou and Hu Chunhua through prior overlap with Hu in Tibet. Hu Chunhua also satisfies the CCYL criterion through his leadership experience in the central CCYL. Together, these two officials correspond to three provincial party-secretary tenures.

reflects incumbent connection in general or is specific to the personalist turn. If connection to the sitting general secretary is sufficient, Hu-connected provincial party secretaries should display a similar selection pattern, promoting city leaders with lower reported GDP performance without corresponding weakness in less directly manipulable indicators of local economic activity. If the pattern is instead specific to the personalist turn rather than a general feature of incumbent connection, it should be more pronounced under Xi-protégé provincial party secretaries than under Hu-connected ones.

Cadre Career Records and Performance Profiles

I construct a leader-level career dataset covering provincial party secretaries, prefectural party secretaries, and mayors who served between January 2002 and December 2024. The dataset records each official's name, birth month, gender, education, office, jurisdiction, and start and end months. A tenure is defined as a continuous spell in a given office. The sample includes all tenures that overlap with the 2002–2024 observation window, including officials who entered office before January 2002 but continued to serve after the beginning of the sample period. Ongoing tenures are right-censored in December 2024.

The provincial party-secretary data cover all 31 provincial-level units during the period under study. The city-leader data cover 334 prefecture-level cities in 27 provincial-level units. After trimming city-leader spells to the 2002–2024 observation window, the city-leader dataset contains 5,121 mayoral and prefectural party-secretary spells held by 3,372 individuals, including 2,602 mayoral spells and 2,519 prefectural party-secretary spells. The records are broadly comprehensive. Out of 7,682 possible city-years in the 334-city panel, 47 lack at least one city-level leader record. These missing cells are concentrated in the earliest years of the panel and in later special cases of administrative change or city creation.

I convert mayoral and prefectural party-secretary spells into an annual city-year-role panel. Each city-year is matched to one mayor and one prefectural party secretary. When turnover occurs within a year, I assign the city-year to the official who served for more than six months. If no official served for more than six months, I assign the city-year to the official with the longest service in that

year. This coding rule follows Yao and Zhang's (2015) city-year leader assignment convention. The resulting annual panel contains 15,297 city-year-role observations, covering 7,662 city-years, 334 cities, and 3,259 city leaders between 2002 and 2024.

The purpose of this panel is to estimate city leaders' revealed economic performance. This is not straightforward because annual economic outcomes are observed at the city-year level, while local leadership is shared by two officials. In Chinese cities, the prefectural party secretary has authority over personnel and major political decisions, while the mayor manages the daily operation of the municipal government. Both can shape local economic performance, but their interaction is difficult to parameterize. I therefore follow Yao and Zhang's (2015) approach of treating mayors and prefectural party secretaries as making separate contributions to city outcomes. The same city-year economic outcome can enter the estimation twice, once for the mayor and once for the prefectural party secretary.

The city-level economic data are drawn from the China City Statistical Yearbooks. These yearbooks report annual city statistics in the following publication year, which helps limit concerns that the measures reflect later historical revisions rather than the information environment available close to the time of evaluation. The baseline outcome is real GDP per capita growth, measured as the annual log change in nominal city-level GDP per capita minus province-level log CPI inflation. Growth rates are calculated only for consecutive city-year observations. Because the yearbooks report hukou-registered population in most years but permanent-resident population in 2020 and 2023, I code population observations in those two years as missing to maintain a consistent denominator. The GDP per capita profile is therefore estimated from available comparable growth observations spanning 2003–2022.

To estimate revealed performance profiles, I adapt Yao and Zhang's (2015) mover-based fixed-effects decomposition approach. Movers—officials who served as mayor or prefectural party secretary in more than one city—connect leaders and localities within a common connected component, making it possible to separate leader-specific performance profiles from city fixed effects. I therefore restrict the estimation to the largest connected component in the leader–city–province–year

network and estimate leader fixed effects using the following specification:

$$y_{icrt} = \theta_i + \psi_c + \lambda_{p(c)t} + \beta_1 \text{InitialLevel}_{icr} + \beta_2 \log(\text{Population}_{ct}) + \varepsilon_{icrt}. \quad (1)$$

Here, y_{icrt} is the annual growth outcome associated with leader i , city c , role r , and year t . θ_i is the leader fixed effect and provides the revealed performance profile used in the selection analysis. ψ_c is a city fixed effect, and $\lambda_{p(c)t}$ is a province-year fixed effect. $\text{InitialLevel}_{icr}$ is the lagged log level of the relevant outcome measured at the beginning of the leader's spell, and $\log(\text{Population}_{ct})$ controls for city population.³

My specification follows Yao and Zhang's (2015) baseline logic, with one adaptation: I replace year fixed effects with province-year fixed effects to account for province-specific annual conditions. This choice fits my setting because city-level outcomes may reflect provincial growth targets, fiscal and credit conditions, policy campaigns, promotion cycles, and statistical practices that vary across provinces and years. Province-year fixed effects absorb common annual environments shared by cities within the same province, while retaining within-province-year variation needed to estimate leader-specific performance profiles.

I also use two alternative indicators of local economic activity to distinguish reported GDP growth from other dimensions of economic expansion that are less directly manipulable through local statistical reporting. The first alternative outcome is real growth in year-end total outstanding loans of financial institutions, measured as the annual log change in nominal loan balances minus province-level log CPI inflation. Credit-related indicators are useful because lending is tied to financial contracts and interest payments, and China's banking system is vertically organized and administratively separate from prefecture-level governments. Major state-owned commercial banks also occupy administrative ranks above most city leaders. City leaders may influence credit conditions through development strategy, public investment, or land-based financing, but they should find it harder to fabricate loan data directly than local GDP statistics.

3. Because log population enters all leader-profile models, the population harmonization described above applies across outcomes. The resulting profile samples end in 2022. Table C.1 in the Supplemental Materials reports the outcome-specific sample periods and sizes.

The second alternative outcome is nighttime-light growth within city administrative boundaries. Nighttime lights are independently recorded by satellites and have been widely used as a proxy for economic activity, especially where official growth statistics are unavailable, weak, or potentially manipulated (Henderson, Storeygard, and Weil 2012; Hu and Yao 2022; Martínez 2022; Zeng and Zhou 2024). I draw on harmonized annual nighttime-light rasters covering 2002 to 2024 (Z. Chen et al. 2026) and extract city-level luminosity using annual prefecture-level boundary maps from the China Township Administrative Boundary dataset (Cheng, Zhang, and Chen 2025). Using annual boundaries is important because prefectural boundaries changed over time; it avoids imposing contemporary administrative boundaries on earlier years. For each city-year, I calculate total nighttime-light intensity within the administrative boundary and then compute its annual log change from consecutive city-year observations. Unlike GDP and loan growth, nighttime-light growth is not adjusted for inflation because luminosity is a physical measure rather than a monetary aggregate.⁴

Across all outcomes, I apply the same basic restrictions. I exclude city-years affected by administrative boundary changes, since boundary changes can mechanically alter GDP, population, loans, and nighttime-light aggregates.⁵ For each direct growth outcome, I trim the dependent variable at the 1st and 99th percentiles before estimating leader effects. After selecting the largest connected component, I iteratively drop singleton fixed-effect groups so that leader, city, and province-year effects are estimated from connected within-sample comparisons. I center the estimated leader effects around their mean and express them in percentage-point units. I refer to these estimates as revealed performance profiles because they capture relative leader-level differences in local growth

4. Table B.1 in the Supplemental Materials reports descriptive statistics for the city-year economic variables used in the analysis, before outcome-specific trimming, connected-component restrictions, and conversion to leader-year-role observations.

5. These include well-known prefecture-level reorganizations such as the 2011 dissolution of Chaohu and the redistribution of its territory across Hefei, Wuhu, and Ma'anshan, and the 2019 absorption of Laiwu into Jinan. I also exclude city-years in which annual boundary maps indicate a year-to-year administrative-area change of at least 5 percent. This rule flags 35 city-years, or 0.46 percent of the city-year master panel; 24 of these city-years are flagged by the 5-percent mapped-area rule.

outcomes within the connected sample.

The final estimating samples vary by outcome: 8,999 leader-year-role observations for GDP per capita growth, 9,501 for real loan growth, and 10,074 for nighttime-light growth. Table C.1 in the Supplemental Materials reports the corresponding fixed-effect models, outcome-specific samples, model fit, and diagnostic comparisons with restricted models that omit leader fixed effects.

Measuring Statistical Restraint in Reported GDP

I measure statistical restraint in reported GDP using a leader-level GDP–nighttime-light (GDP–NTL) residual profile. Reported GDP growth may reflect both real economic activity and politically motivated statistical overstatement. I do not assume that nighttime lights measure true GDP directly. Rather, because nighttime lights are independently recorded by satellites and are less directly manipulable by city officials, I use them as an externally observed proxy for local economic activity. The goal is to construct a benchmarked reported GDP measure that compares officials in terms of their reported GDP performance relative to an externally observed activity proxy and contextual fixed effects. This benchmarking logic builds on studies that use nighttime lights and other external indicators to assess the credibility of official GDP statistics and detect GDP overstatement or statistical management (e.g., Henderson, Storeygard, and Weil 2012; Hu and Yao 2022; Lyu et al. 2018; Martínez 2022; Wallace 2016; Zeng and Zhou 2024).

To construct the GDP–NTL residual profile, I proceed in two steps. First, using the cleaned city-year panel described above, I apply the same boundary-change exclusions and trim real GDP per capita growth at the 1st and 99th percentiles. I then estimate the following first-stage residualization model:

$$g_{ct}^{\text{GDP}} = \alpha_c + \delta_t + \rho g_{ct}^{\text{NTL}} + u_{ct}. \quad (2)$$

Here, g_{ct}^{GDP} is real GDP per capita growth in city c and year t , and g_{ct}^{NTL} is boundary-level nighttime-light growth. α_c and δ_t are city and year fixed effects. The residual u_{ct} is the city-year residual from this benchmark model after absorbing persistent city differences, common annual shocks, and the fitted association with nighttime-light growth. Positive values indicate higher reported

growth relative to the nighttime-light benchmark. The first-stage sample contains 4,874 city-year observations from 289 cities between 2003 and 2022. Column (4) of Table C.1 and Table C.2 in the Supplemental Materials report the leader-level fixed-effect model and the first-stage residualization model, respectively. Table C.3 reports diagnostic specifications for the GDP–NTL association.

I then convert this city-year residual into a leader-level profile using the same mover-based fixed-effects framework described above. Because each city-year is linked to both the mayor and the prefectural party secretary, the residualized outcome enters the leader-year-role panel once for each officeholder:

$$\widehat{u}_{icrt} = \kappa_i + \psi_c + \lambda_{p(c)t} + \beta_1 \text{InitialGDPpc}_{icr} + \beta_2 \log(\text{Population}_{ct}) + \varepsilon_{icrt}. \quad (3)$$

Here, $\widehat{u}_{icrt}$ is the residualized GDP-growth outcome associated with leader i , city c , role r , and year t . κ_i is the leader fixed effect and provides the leader-level GDP–NTL residual profile. ψ_c is a city fixed effect, and $\lambda_{p(c)t}$ is a province-year fixed effect. $\text{InitialGDPpc}_{icr}$ is the initial lagged log GDP per capita for the leader-spell, and $\log(\text{Population}_{ct})$ controls for city population.

I center the estimated leader fixed effects around their mean and multiply them by 100, so the resulting profile is expressed in approximately percentage-point units. Higher values indicate that a leader is more strongly associated with higher reported GDP growth relative to the nighttime-light benchmark, while lower values indicate greater revealed statistical restraint in reported GDP. Because nighttime lights are externally observed and less directly manipulable by city officials, this profile provides a comparable, externally benchmarked measure for assessing leader-level differences in reported GDP performance. The final GDP–NTL residual profile is estimated from 8,999 leader-year-role observations, covering 1,842 leaders and 283 cities. Table D.1 in the Supplemental Materials summarizes the distributions of the estimated leader-level profiles.

Promotion Attribution and Analysis Sample

Following Jiang’s (2018) promotion-based approach, I identify a city leader as selected by the provincial party secretary in office when the official was first promoted to mayor or prefectural party secretary. Rather than inferring ties from shared backgrounds or overlapping work histo-

ries, this approach treats first-time appointment into a key city leadership post as evidence of a politically consequential personnel relationship. Recent work on factional identification describes Jiang’s within-tenure promotion approach as one of the most accessible and justifiable strategies for detecting patronage ties (Zhang et al. 2025). My use of this method is narrower: I use the promotion event to attribute selection responsibility, not to claim that every promoted official is necessarily a personal client.

The selection dataset is organized at the role-specific first-appointment level. Each observation is an official’s first appointment as mayor or prefectural party secretary, matched to that official’s estimated leader-level revealed performance profile. An official may therefore enter the dataset once for each role. Subsequent transfers or rotations within the same role are not treated as additional selection events. To handle provincial leadership transitions, I apply a three-month buffer: first appointments to city-leader posts made during the incoming provincial party secretary’s first three months in office are attributed to the predecessor.

The main selection analyses cover first-time city-leader appointments made between 2004 and 2019. The starting year allows the analysis to include the Hu-era benchmark, while the ending year reflects the coverage structure of the estimated leader-level profiles. Figure D.1 in the Supplemental Materials reports annual profile match rates by leader-spell start year. For the GDP–NTL residual profile, the match rate exceeds 75 percent from 2004 to 2016 and remains close to that level in 2017 at 74.4 percent. It declines to 67.7 percent in 2018 and 47.1 percent in 2019, before falling sharply to 27.8 percent in 2020. This decline is expected because officials appointed near the end of the observation window have fewer subsequent city-year performance observations and fewer observed cross-city moves linking them to the mover-based profile estimates. I therefore end the main profile-based selection sample in 2019, before profile coverage deteriorates sharply. Substantively, this window also captures the main pre-2020 phase of Xi-era power consolidation, including the period leading up to and following the 2018 abolition of presidential term limits.

I divide the sample into two periods. The Hu-era benchmark covers first-time city-leader appointments from 2004 to 2013, and the Xi-era analysis covers those from 2014 to 2019. I exclude

appointments that do not correspond to ordinary prefecture-level personnel authority, including provincial capitals and sub-provincial cities, where top leadership posts often carry vice-provincial rank and fall outside the ordinary personnel authority of provincial party secretaries. In the prefectural party secretary sample, I also exclude officials who had already reached vice-provincial rank before becoming prefectural party secretary.

After these restrictions, the 2004–2019 selection sample contains 2,409 first-time city-leader appointments before outcome-specific profile missingness, including 1,306 mayor appointments and 1,103 prefectural party-secretary appointments. The Hu-era benchmark includes 1,472 appointments: 416 under Hu-connected provincial party secretaries and 1,056 under non-connected provincial party secretaries. The Xi-era analysis includes 937 appointments: 157 under Xi-protégé provincial party secretaries and 780 under non-protégé provincial party secretaries. Tables E.1 and E.2 in the Supplemental Materials report the composition of the selection sample and the group means of the four revealed-performance profiles by era and promoter type, respectively.

Main Results

Incompetence or Restraint?

I begin with within-era comparisons in the Xi and Hu periods. In each period, I compare city leaders promoted under incumbent-connected provincial party secretaries with those promoted under non-connected provincial party secretaries:

$$Y_{iptr}^k = \beta_e \text{Connected}_{iptr} + \alpha_p + \lambda_t + \rho_r + \varepsilon_{iptr}, \quad e \in \{\text{Hu}, \text{Xi}\}. \quad (4)$$

where Y_{iptr}^k is one of the four revealed-performance profiles for official i , promoted in province p , year t , and role r . Connected_{iptr} indicates whether the promoting provincial party secretary is connected to the incumbent national leader under the three-month buffer rule. In the Hu-era models, this variable identifies Hu-connected provincial party secretaries; in the Xi-era models, it identifies Xi-protégé provincial party secretaries. The fixed effects α_p , λ_t , and ρ_r denote province, promotion-year, and role fixed effects. Standard errors are clustered by province, and wild-cluster bootstrap p-values are reported in the appendix tables (Cameron, Gelbach, and Miller 2008).

Panel A of Figure 2 reports the Xi-era estimates. The coefficient for reported real GDP per capita growth is negative, indicating that city leaders promoted under Xi-protégé provincial party secretaries are associated with lower reported GDP performance than those promoted under non-protégé provincial party secretaries. Taken alone, this result could be read as evidence that Xi protégés select economically weaker officials. The loan-growth and nighttime-light estimates, however, help distinguish this interpretation from the selection-for-restraint interpretation. The loan-growth estimate is positive, and the nighttime-light estimate is close to zero. Neither result supports the view that Xi-protégé provincial party secretaries systematically select officials who are weaker across broader measures of local economic activity.

The GDP–NTL residual profile points to a different interpretation. City leaders promoted under Xi-protégé provincial party secretaries score about 1.1 percentage points lower on the GDP–NTL residual profile than those promoted under non-protégé provincial party secretaries, roughly 0.22 standard deviations relative to the full distribution of the GDP–NTL residual profile reported in Table D.1. Because this profile measures reported GDP growth relative to satellite-observed nighttime-light growth after accounting for city and temporal fixed effects, lower values indicate lower reported GDP performance relative to an external activity benchmark. This pattern suggests that the negative GDP result does not simply reflect uniformly weaker underlying economic activity. Rather, Xi protégés appear to select officials whose reported GDP performance is lower relative to an externally observed benchmark. Columns (1)–(4) of Table F.1 in the Supplemental Materials report the full estimates for the four revealed-performance profiles. Columns (5)–(8) show that the results remain substantively unchanged after adding individual-level controls for gender, education level, and age at appointment.

Panel B of Figure 2 reports the corresponding Hu-era estimates. Unlike the Xi-era results, the Hu-era estimates are generally small and do not show a comparable negative pattern in either reported GDP performance or the GDP–NTL residual profile. Table F.2 reports the full Hu-era estimates, which remain substantively similar after adding individual-level controls. This contrast suggests that the Xi-era pattern is not a general feature of incumbent-connected provincial party

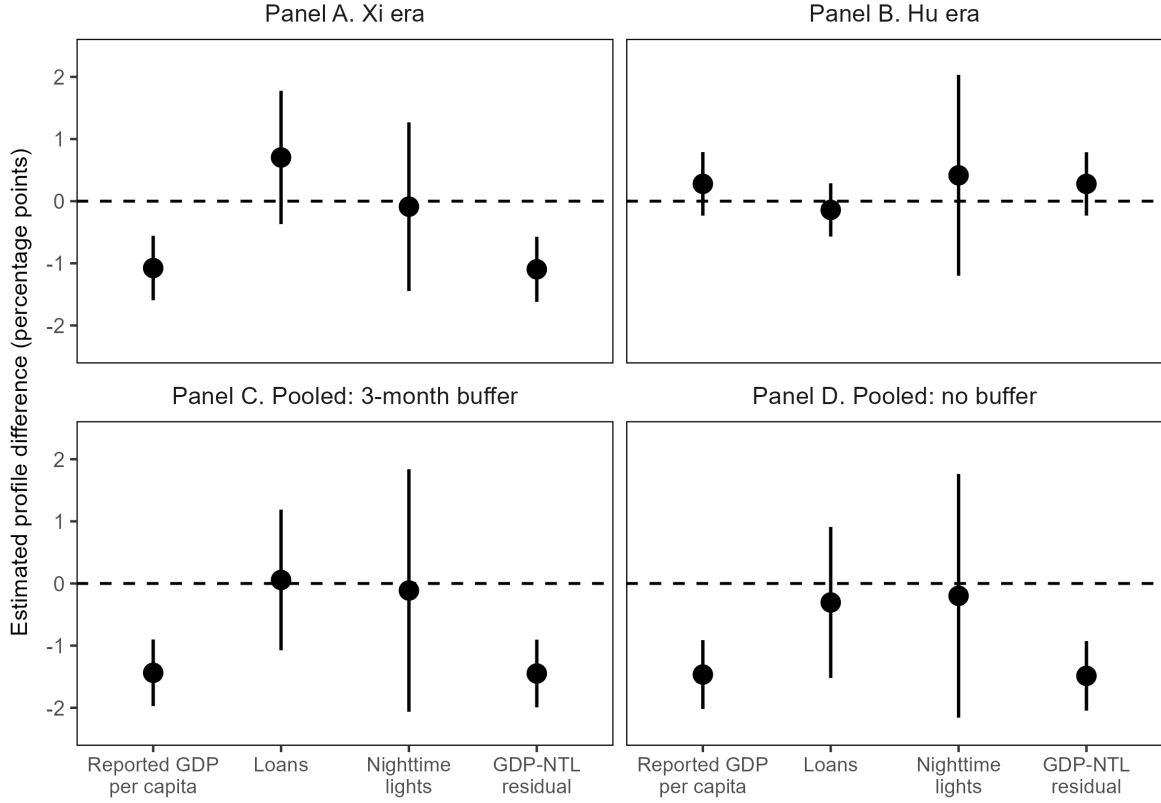


Figure 2: Selection Gaps across Revealed Economic Performance Profiles

Notes: Points report coefficient estimates from the main selection models. Panel A reports the Xi-era within-era gap between city leaders promoted under Xi-protégé and non-protégé provincial party secretaries, using the three-month buffer coding. Panel B reports the corresponding Hu-era gap between city leaders promoted under Hu-connected and non-connected provincial party secretaries. Panels C and D report interaction terms from pooled Hu–Xi benchmark models, capturing the additional Xi-era protégé gap relative to the Hu-era connected benchmark, using the three-month buffer coding and no-buffer coding, respectively. Bars show 95% confidence intervals based on province-clustered standard errors. Profile values are centered leader fixed effects multiplied by 100 and are expressed in approximately percentage-point units. The corresponding regression estimates are reported in columns (1)–(4) of Tables F.1–F.4, respectively.

secretaries, but is instead concentrated among city leaders promoted under Xi-protégé provincial party secretaries.

To assess the Xi–Hu contrast more directly, I next estimate pooled Hu–Xi benchmark models. The within-era estimates show the connection gap separately in each period, while the pooled

models test whether the Xi-era connection gap differs from the corresponding Hu-era benchmark:

$$Y_{iptr}^k = \beta \text{Connected}_{iptr} + \delta Xi_t + \theta (\text{Connected}_{iptr} \times Xi_t) + \alpha_p + \lambda_t + \rho_r + \varepsilon_{iptr}. \quad (5)$$

In this specification, Xi_t indicates promotions in the Xi era. Because the model includes promotion-year fixed effects, the main effect of Xi_t is absorbed by λ_t and is not separately interpreted. The coefficient β captures the Hu-era connected-versus-nonconnected difference, while θ captures the additional Xi-era difference relative to the Hu-era benchmark. The goal of this pooled benchmark comparison is not to estimate a treatment effect from within-unit changes over time, but to ask whether the selection pattern under Xi-protégé provincial party secretaries differs from the Hu-era incumbent-connection benchmark. I estimate this model using both the three-month buffer coding and a no-buffer version of the connection measure.

Panels C and D of Figure 2 report the pooled benchmark results. Tables F.3 and F.4 report the corresponding full estimates under the three-month-buffer and no-buffer coding, respectively. They are consistent with the within-era estimates. The additional Xi-era connection gap is negative for reported GDP per capita growth and, more clearly, for the GDP–NTL residual profile. Depending on whether the three-month buffer is used, the pooled estimates imply an additional Xi-era protégé gap of about 1.5 percentage points on the GDP–NTL residual profile. By contrast, the pooled estimates for loan growth and nighttime-light growth do not show a comparable negative pattern. This reinforces the main interpretation: city leaders promoted under Xi-protégé provincial party secretaries are not uniformly weaker across indicators of local economic activity. Rather, the negative gap is concentrated in the reported GDP growth and GDP–NTL residual profiles, consistent with greater statistical restraint.⁶

Role Heterogeneity: Party Secretaries versus Mayors

I next examine whether the selection-for-restraint pattern differs between mayor promotions and prefectural party-secretary promotions. This distinction is substantively important because the two

6. The number of observations differs across outcomes because not every first-time appointment can be matched to every revealed-performance profile.

promotions differ in how much information provincial party secretaries are likely to have about the promoted official. Mayor promotions often represent an official's first entry into a major city-level executive post. By contrast, prefectural party-secretary promotions more often occur after officials have accumulated prior local executive experience, including mayoral service. Their records are therefore more observable before promotion. If Xi-protégé provincial party secretaries select for restraint, this pattern should be especially visible when they promote prefectural party secretaries.

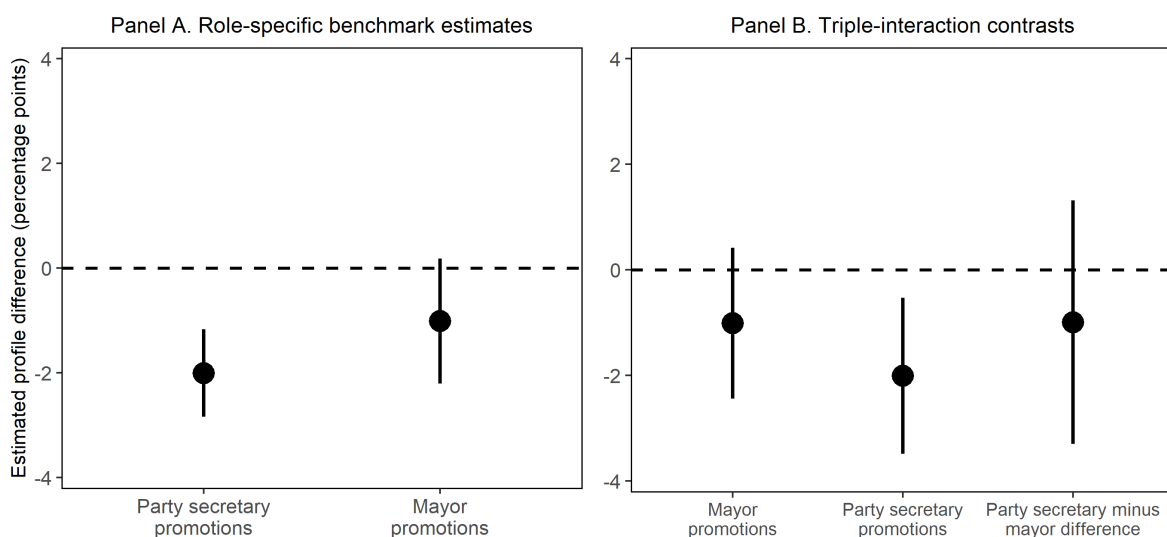


Figure 3: Role Heterogeneity in Selection for Statistical Restraint

Notes: Points report estimated differences in the GDP–NTL residual profile. Panel A reports role-specific pooled Hu–Xi benchmark estimates; each point is the additional Xi-era protégé gap relative to the Hu-era connected benchmark within the corresponding role-specific sample. Panel B reports interpretable contrasts from the triple-interaction role-heterogeneity model. Bars show 95% confidence intervals based on province-clustered standard errors in Panel A and CR2 standard errors clustered by province in Panel B. More negative values indicate lower reported GDP performance relative to the nighttime-light benchmark. The corresponding regression results are reported in Tables G.1 and G.2 for Panels A and B, respectively.

Figure 3 examines role heterogeneity using the GDP–NTL residual profile. Panel A estimates the pooled Hu–Xi benchmark model separately for prefectural party-secretary promotions and mayor promotions. The additional Xi-era protégé gap is descriptively larger among prefectural party-secretary promotions than among mayor promotions. The estimate is -2.0 percentage points on the GDP–NTL residual profile for prefectural party-secretary promotions, compared with -1.0

percentage points for mayor promotions.⁷

To assess this role difference more directly, I estimate a pooled role-heterogeneity model that interacts incumbent connection, the Xi era, and an indicator for prefectural party-secretary promotions. The model includes province-by-role and year-by-role fixed effects, allowing provincial and temporal patterns to differ across the two promotion types. Panel B reports the corresponding contrasts. The ordering remains the same: the additional Xi-era protégé gap is more negative for prefectural party-secretary promotions than for mayor promotions, although the difference between the two role-specific gaps is not statistically distinguishable from zero.⁸

Addressing Alternative Explanations

Several alternative explanations warrant attention. First, the results could be affected by differential coverage across the revealed-performance profiles. The four profiles are not observed for exactly the same set of first-time city-leader appointments. Table E.3 in the Supplemental Materials shows that appointments under Xi-protégé provincial party secretaries are less likely to be matched to the loan-growth and nighttime-light profiles, while match coverage does not differ meaningfully for the GDP-growth or GDP–NTL residual profiles. To address any remaining concern that differences in sample composition drive the cross-outcome pattern, I re-estimate the Xi-era selection models on the common sample of appointments that can be matched to all four profiles. Table F.5 shows that the same substantive pattern remains, with the negative Xi-protégé gap concentrated in the GDP-growth and GDP–NTL residual profiles rather than in loan growth or nighttime-light growth.

Second, an important concern is that the results could reflect broader differences between the Xi and Hu eras rather than a selection pattern specific to Xi-protégé provincial party secretaries. The Xi era coincided with more intensive campaign-style governance initiatives, including anticorruption, poverty alleviation, and environmental enforcement (Jia and Chen 2019; Smith 2018; Wang 2022), each of which could have pushed local officials toward more restrained governance. I ad-

7. Table G.1 in the Supplemental Materials reports the corresponding role-specific pooled benchmark estimates.

8. Table G.2 in the Supplemental Materials reports the triple-interaction model and the corresponding contrasts.

dress this concern with the two diagnostic models reported in columns (1) and (2) of Table H.1 in the Supplemental Materials. The first restricts the sample to appointments made under non-connected provincial party secretaries in the Hu era and non-protégé provincial party secretaries in the Xi era. This model tests whether Xi-era appointments generally exhibit lower GDP–NTL residual profiles even among officials not selected by connected or protégé provincial party secretaries. The second model uses the pooled Hu–Xi sample to separate three components: the general Xi-era shift among appointments made under non-connected or non-protégé promoters, the Hu-era connected-promoter gap, and the additional Xi-era protégé gap relative to the Hu-era benchmark. Because these models estimate period-level differences directly, they replace promotion-year fixed effects with a centered linear promotion-month trend while retaining province and role fixed effects. The results do not show a comparable negative shift among appointments made under non-connected or non-protégé provincial party secretaries. The negative difference instead remains concentrated in the additional Xi-era protégé gap. This pattern makes it less likely that the main result is simply a byproduct of Xi-era policy campaigns or broader period effects.

This diagnostic does not rule out the possibility that Xi-era campaigns operated differently through Xi-protégé provincial party secretaries. The leader-profile estimates include province-year fixed effects, which reduce concern that estimated profiles are driven by province-specific annual shocks common to cities within the same province-year. This structure also narrows the campaign-based alternative. If Xi-protégé provincial party secretaries simply imposed uniform statistical discipline on all cities in their jurisdictions, that common province-year component would be absorbed by the fixed effects rather than loaded onto leader profiles. The estimated gaps therefore cannot be generated by uniform provincial pressure alone; they require within-province-year differences across officials. However, the design cannot fully rule out targeted statistical discipline applied differentially to selected officials, or stronger anticipation of central discipline among the officials protégés choose. Because the key outcome in the main analysis is based on reported GDP, the profile-based evidence alone cannot resolve this narrower possibility. The pre-selection analysis below therefore examines prior mayoral spells and outcomes beyond reported GDP to assess

whether later Xi-protégé appointees had already displayed more restrained governing trajectories before promotion.

Third, the lower GDP–NTL residual profiles could reflect differences in the local economic environments in which Xi-protégé provincial party secretaries operate. If Xi protégés governed provinces with more favorable economic conditions, or if the city leaders they promoted were assigned to cities with stronger pre-promotion economic trajectories, these officials might have faced less pressure to inflate reported GDP figures. The main specifications include province fixed effects, which absorb time-invariant differences across provinces. I further address the city-level version of this concern in column (3) of Table H.1 by controlling for average nighttime-light growth in the three years before promotion. The coefficient on pre-promotion nighttime-light growth is positive and statistically significant, suggesting that stronger prior local activity is associated with higher, not lower, GDP–NTL residual profiles. More importantly, the Xi-era protégé gap remains negative after this adjustment, indicating that the main result is not explained by observable pre-promotion economic trajectories in the cities where these officials served.

Fourth, the results could be driven by a single influential province, especially given well-known cases of local GDP manipulation such as Liaoning, or by a single coded Xi-protégé provincial party secretary. The latter concern is particularly relevant because the Xi-protégé coding includes Li Xi as a separately justified literature-based case in addition to officials identified through the main hierarchical-overlap rule. To address these concerns, Figures I.1–I.6 in the Supplemental Materials report province- and protégé-level influence diagnostics for three key specifications: the pooled Hu–Xi benchmark model, the model restricted to prefectural party-secretary promotions, and the Xi-era model with the pre-promotion nighttime-light control. Figures I.1–I.3 re-estimate these models after omitting one province at a time, while Figures I.4–I.6 repeat the exercise after omitting one Xi-protégé provincial party secretary at a time from the corresponding model sample.⁹ Across both

9. The leave-one-protégé-out diagnostics are estimated within the relevant outcome- and role-specific model samples. The main 2014–2019 Xi-era selection sample includes 10 Xi-protégé provincial party secretaries before outcome-specific profile matching. For the GDP–NTL residual-profile diagnostics, the pooled benchmark and Xi-era pre-promotion-NTL specifications include eight Xi-protégé provincial party secretaries with matched appointments. The party-secretary-

sets of exclusions, the key estimates remain negative and close to the full-sample estimates, including when Liaoning is omitted and when Li Xi or any other Xi-protégé provincial party secretary represented in the relevant model sample is excluded.

Fifth, the results might depend on using nighttime lights as the external benchmark for reported GDP. Although the loan-growth profile provides a separate, less directly manipulable indicator of local economic activity, it does not directly address whether the residual-based results depend specifically on the nighttime-light benchmark. To address this benchmark-specific concern, I replicate the selection analysis using city-level total electricity consumption, a physical activity indicator that is harder for prefectural officials to manipulate directly than reported GDP (W. Chen et al. 2019; Wallace 2016). I estimate both an electricity-consumption growth profile and a GDP–electricity residual profile, applying the same boundary-change exclusions, trimming rules, and fixed-effect structure as in the main performance-profile analyses. Tables J.1 and J.2 in the Supplemental Materials report the residualization and profile-estimation models, respectively, and Table J.3 reports the corresponding selection models. The results reinforce the main interpretation. The electricity-consumption growth estimates do not show a comparable negative pattern for city leaders promoted under Xi-protégé provincial party secretaries. After benchmarking reported GDP growth against electricity-consumption growth, however, Xi-protégé appointees have significantly lower GDP–electricity residual profiles in both the Xi-era and pooled benchmark specifications. This pattern is consistent with restraint concentrated in reported performance.

Pre-Selection Evidence

The preceding results are consistent with the argument that personalist rule reshapes local political selection through the ruler’s protégés. Xi-protégé provincial party secretaries promote city leaders with lower GDP–NTL residual profiles, while these appointees do not appear systematically weaker on loan growth or nighttime-light growth. Yet two related concerns remain. First, because leader-level performance profiles are estimated from observed behavior in office, lower GDP–NTL residual

promotion diagnostic includes seven because one of these protégés has no matched first appointment as prefectural party secretary in the role-specific sample.

profiles may partly reflect behavior under stricter post-selection statistical discipline rather than a pre-existing governing style. Second, Xi-protégé provincial party secretaries may have been especially likely to enforce or anticipate central disciplinary pressure against inflated GDP reporting. If so, the lower GDP–NTL residual profiles documented above could reflect a narrower response to statistical discipline rather than a broader selection logic favoring restraint. I address these concerns by moving one step earlier in the career sequence. I examine the most recent prior mayoral spells of officials who later became first-time prefectural party secretaries and assess whether they had already displayed restraint in domains beyond reported GDP.

The start of a mayoral spell is a useful setting because city leadership turnover is a critical moment for local policy choice (Chen and Zhang 2021; Guo 2009; Liu 2026; Shen, Wu, and Wu 2022). Although mayors do not determine local economic policy alone, mayoral entry provides a clear leadership-transition point for observing whether incoming officials followed different governing trajectories. The analysis covers officials who were promoted to prefectural party secretary for the first time between 2014 and 2024. Unlike the main selection models, this pre-selection analysis does not require matching later party-secretary spells to leader-level performance profiles, allowing the sample to extend beyond 2019. For each promotion, I identify the official’s most recent prior mayoral spell and define the later-selected group according to whether the later first-time prefectural party-secretary appointment was made under a Xi-protégé provincial party secretary, using the same three-month attribution rule as in the main analysis. I exclude prior mayoral spells that began under Xi-protégé provincial party secretaries to ensure that the observed mayoral behavior was not already shaped by an earlier Xi-protégé appointment. The resulting base sample contains 410 prior mayoral spells, including 70 events in the later Xi-protégé-selected group attributed to 16 Xi-protégé provincial party secretaries holding 18 provincial party-secretary tenures and 340 comparison events attributed to 71 non-protégé provincial party secretaries holding 77 provincial party-secretary tenures. This design therefore functions as a pre-selection diagnostic: it compares the prior governing behavior of officials later selected by Xi protégés with that of officials later selected by non-protégé provincial leaders. If officials later selected by Xi-protégé provincial party

secretaries already exhibited more restrained governing trajectories as mayors, then restraint was visible before the later party-secretary promotion rather than emerging only after selection.

I focus on two observable domains in which politically cautious governance should be visible. The first is fiscal expenditure growth. In China's promotion-oriented local government system, fiscal expansion is not merely a neutral budgetary outcome but a governing instrument through which local officials finance visible projects, stimulate investment, and generate short-term political achievements (Guo 2009; Jia, Guo, and Zhang 2014; Que, Zhang, and Schulze 2019). Lower fiscal expenditure growth therefore indicates a more restrained fiscal style, reflecting less reliance on rapid public-spending expansion as a strategy of local development.

The second indicator is land-transfer revenue expansion.¹⁰ Land finance is central to China's local developmental model. Local governments use land conversion, land leasing, and real estate-related development to finance infrastructure, support urban construction, and generate extra-budgetary revenue (Ding 2003; Ong 2014; Shu et al. 2018; Wu, Li, and Yan 2015). At the same time, aggressive land-based development can create political and fiscal risks, including debt exposure, real estate dependence, land expropriation conflicts, and blame from destabilizing local development. Because land-transfer revenue is highly volatile and sensitive to low base values, I use an expansion indicator equal to one if city-level land-transfer revenue increases relative to the previous year. Lower land-transfer revenue expansion therefore captures restraint in a domain where local officials can pursue rapid growth but may also expose their superiors to political risk.

I estimate stacked event-study models around the start of the prior mayor spell:

$$Y_{it} = \sum_{k \neq -1} \beta_k [T_i \times 1\{\text{EventTime}_{it} = k\}] + \mu_k + \alpha_i + \lambda_t + \varepsilon_{it}. \quad (6)$$

Here, Y_{it} is the outcome for event i in calendar year t . T_i is an indicator equal to one if the official was later promoted to prefectural party secretary under a Xi-protégé provincial party secretary. EventTime_{it} measures years relative to the start of the prior mayor spell, with event time -1 as the omitted category. Event-time fixed effects, μ_k , absorb common changes around mayoral entry

10. Land-transfer revenue is compiled from land-use-rights conveyance transaction records on the China Land Market Network (中国土地市场网, <https://www.landchina.com>).

across prior mayoral spells, while event fixed effects, α_i , absorb time-invariant differences across spells. Calendar-year fixed effects, λ_t , absorb common annual shocks. The coefficients β_k estimate the difference between officials later selected by Xi-protégé provincial party secretaries and those later selected by non-protégé provincial party secretaries at event time k , relative to the corresponding difference in the year immediately before mayoral entry. Standard errors are clustered by the prior mayor city.

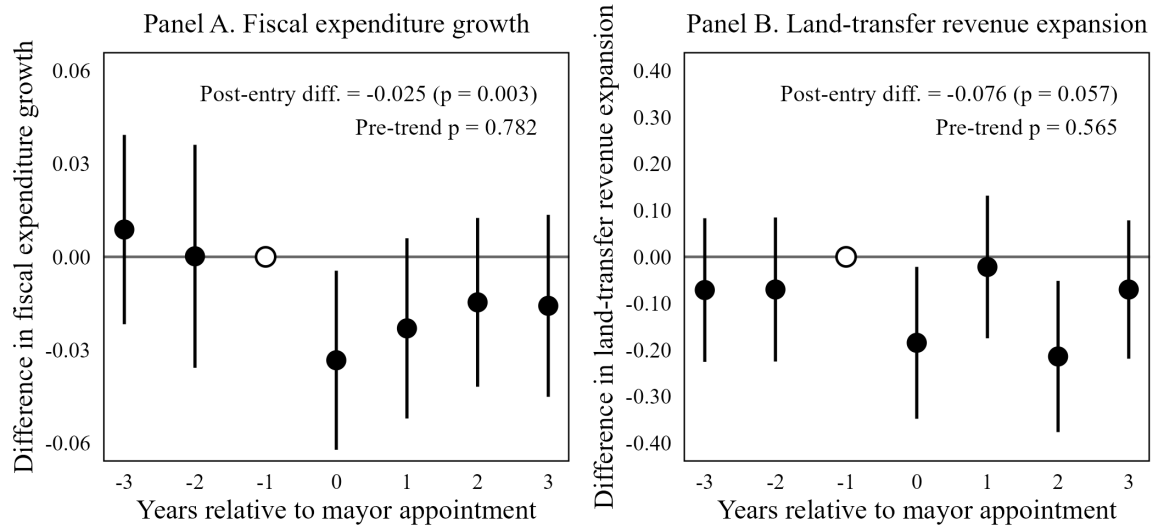


Figure 4: Pre-Selection Evidence from Prior Mayoral Spells

Notes: The figure reports stacked event-study estimates around the start of officials' prior mayoral spells using a symmetric $[-3, +3]$ event window. Panel A reports fiscal expenditure growth, measured as the annual log change in city-level fiscal expenditure. Panel B reports land-transfer revenue expansion, an indicator for whether city-level land-transfer revenue increases relative to the previous year. Points indicate coefficient estimates, and bars indicate 95% confidence intervals. Event time -1 is the omitted reference period; it is normalized to zero and shown as an open marker without a confidence interval. The later-selected indicator equals one if the official was later promoted to prefectural party secretary under a Xi-protégé provincial party secretary. The post-entry difference reports the average gap after mayoral entry between officials later promoted under Xi-protégé provincial party secretaries and comparable later appointees, relative to the average pre-entry difference. The joint pre-trend test reports the joint significance of the lead coefficients. Standard errors are clustered by the city of the prior mayoral spell. The corresponding full estimates are reported in Tables K.1 and K.2 in the Supplemental Materials for Panels A and B, respectively.

Figure 4 reports the stacked event-study estimates for fiscal expenditure growth and land-transfer revenue expansion using a symmetric $[-3, +3]$ event window around mayoral entry. I

use this window as the main specification because it captures officials' early governing trajectories over a substantively meaningful portion of a city-leader spell: prior work on Chinese city leaders reports a median tenure of about three years, despite a formal five-year term (Yao and Zhang 2015). The window also limits exposure to later turnover, subsequent promotion dynamics, and longer-run city-specific shocks.

Panel A shows that, relative to comparable later appointees, officials later promoted under Xi-protégé provincial party secretaries exhibit lower fiscal expenditure growth after entering their prior mayoral posts. The corresponding post-entry estimate of -0.025 indicates that, relative to the comparison group, their annual fiscal expenditure growth shifted downward by roughly 2.5 percentage points from the pre-entry to the post-entry period.¹¹ The divergence emerges most sharply in the year of mayoral entry, when the relative gap widens by approximately 3.3 percentage points compared with the year immediately before entry. Because this is the point at which the incoming mayor begins to exercise executive authority, the timing is consistent with an immediate shift toward a more restrained fiscal governing trajectory rather than a gradually emerging difference. The pre-entry estimates remain close to zero, and the joint pre-trend test provides little evidence of differential trends before mayoral entry.

Panel B reports a similar pattern for land-transfer revenue expansion. Officials later promoted under Xi-protégé provincial party secretaries were less likely than comparable officials to preside over increases in land-transfer revenue during their prior mayoral spells. The corresponding post-entry estimate of -0.076 indicates that the change in the probability of land-revenue expansion from the pre-entry to the post-entry period was 7.6 percentage points lower for these officials than for the comparison group. The event-time estimates are less uniform than those for fiscal expenditure, but the negative differences are especially pronounced in the year of mayoral entry and in the second year thereafter. The pre-entry estimates again provide little evidence of differential trends before

11. As a back-of-the-envelope magnitude, Table B.1 reports mean city-year fiscal expenditure of 28.82 billion yuan. Evaluated at this mean, a 2.5-percentage-point lower annual growth rate corresponds mechanically to about 0.72 billion yuan in lower annual expenditure growth, or roughly 100 million USD using an exchange rate of 7 yuan per dollar.

mayoral entry.

Taken together, these results provide pre-selection evidence that officials later promoted under Xi-protégé provincial party secretaries had already followed more restrained local-development trajectories in domains beyond reported GDP during their prior mayoral spells. Tables K.1 and K.2 in the Supplemental Materials report the full event-time estimates and show that the results are substantively similar in the wider $[-4, +4]$ event window.

Conclusion

Existing accounts of personalization emphasize how rulers weaken constraints, elevate loyalists, and centralize control over appointment and punishment. This article instead argues that personalist rule reshapes political selection beyond the ruler's immediate circle. Once loyalists acquire authority within the hierarchy, they become selectors in their own right. Because their authority depends on the ruler's continued trust, protégés face their own problem of controlled delegation. They need subordinates who can implement priorities and maintain order, but whose governing styles remain controllable and unlikely to generate difficult-to-contain risks.

Evidence from China under Xi supports this argument. Xi-protégé provincial party secretaries do not appear to select simply for less capable officials. Their appointees have lower reported GDP growth profiles and lower GDP–nighttime-light residual profiles, while this negative pattern is not mirrored in less directly manipulable indicators of local economic activity, such as loan growth or nighttime-light growth. Pre-selection evidence from earlier mayoral spells further shows that these later appointees had already followed more restrained fiscal and land-based development trajectories before their later promotion. Taken together, these patterns are more consistent with selection for restraint than with a simple loyalty–competence trade-off.

The indicators analyzed here capture selected, observable manifestations of restraint in local development and performance reporting. They do not exhaust the full range of governing styles that superiors may observe through cadre evaluations, inspection reports, informal reputations, prior implementation records, or repeated interactions within the hierarchy. Even so, the design identifies

a pattern of delegated selection rather than the full motivational process behind it. It can identify protégés' structural vulnerability as a position within the personalist hierarchy, but it cannot directly observe the subjective motives through which individual protégés interpreted that position. I therefore treat vulnerability as the structural logic that makes this selection pattern intelligible, rather than as a separately identified motive.

The broader implication is not that all personalist regimes converge on a single organizational form. Personalist regimes vary across institutions, elite coalitions, coercive structures, and economic settings, and they are not necessarily governed by protégés alone. The empirical analysis focuses on civilian provincial party secretaries, an especially observable channel because they exercise formal authority over prefecture-level appointments. Other trusted agents, including those in military, security, central party, or strategic state-sector positions, may transmit personalist authority through different personnel channels.

Yet this variation should not obscure a common organizational move. Personalist rulers often weaken autonomous elites by placing loyal agents in positions that control careers, appointments, and access to office (Baturu and Elkink 2016; Geddes, Wright, and Frantz 2018; Márquez 2024; Svolik 2012). This article shows that the consequences of that move extend beyond the ruler's inner circle. Once loyal agents acquire personnel authority, their choices shape how personalist power is reproduced below the top leadership. Although the evidence comes from China under Xi, the argument speaks to a broader organizational feature of personalist rule. Personalization changes not only who rules near the top, but also which kinds of officials are selected and empowered below.

These findings also broaden how we understand the governance costs of personalism. Existing work has emphasized more visible consequences of personalist insecurity, including repression, elite purges, coup and assassination risks, succession uncertainty, and elite defection (Chin et al. 2022; Frantz et al. 2020; Grundholm 2020; Kuehn, Llanos, and Richter 2026; Sudduth 2017). The evidence here points to a quieter organizational cost inside the personnel system. By empowering trusted agents as downstream selectors, personalist rule may improve hierarchical control while changing which forms of performance become politically acceptable. Officials associated with

high-variance, politically exposed strategies of performance production are discounted, whereas more cautious and defensible governing styles are favored. The result is not necessarily a simple decline in competence, but a narrowing of bureaucratic initiative away from local developmental entrepreneurship and toward politically legible restraint.

References

- Baturo, Alexander, and Johan A. Elkink. 2016. "Dynamics of Regime Personalization and Patron–Client Networks in Russia, 1999–2014." *Post-Soviet Affairs* 32, no. 1 (January 2, 2016): 75–98. <https://doi.org/10.1080/1060586X.2015.1032532>.
- Baturo, Alexander, Nikita Khokhlov, and Jakob Tolstrup. 2025. "Playing the Sycophant Card: The Logic and Consequences of Professing Loyalty to the Autocrat." *American Journal of Political Science* 69, no. 3 (July 1, 2025): 1180–1195. <https://doi.org/10.1111/ajps.12909>.
- Besley, Timothy, and Masayuki Kudamatsu. 2007. "Making Autocracy Work." *LSE STICERD Research Paper*.
- Blattman, Christopher, Scott Gehlbach, and Zeyang Yu. 2025. *The Personalist Penalty: Varieties of Autocracy and Economic Growth*. National Bureau of Economic Research. <https://doi.org/10.3386/w34093>.
- Bokobza, Laure, Suthan Krishnarajan, Jacob Nyrup, Casper Sakstrup, and Lasse Aaskoven. 2022. "The Morning After: Cabinet Instability and the Purging of Ministers after Failed Coup Attempts in Autocracies." *The Journal of Politics* 84, no. 3 (July 1, 2022): 1437–1452. <https://doi.org/10.1086/716952>.
- Brødsgaard, Kjeld Erik. 2018. "China's Political Order under Xi Jinping: Concepts and Perspectives." *China: An International Journal* 16 (3): 1–17.
- Bueno de Mesquita, Bruce, Alastair Smith, Randolph M. Siverson, and James D. Morrow. 2003. *The Logic of Political Survival*. MIT Press.
- Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller. 2008. "Bootstrap-Based Improvements for Inference with Clustered Errors." *The Review of Economics and Statistics* 90, no. 3 (August 1, 2008): 414–427. <https://doi.org/10.1162/rest.90.3.414>.

- Chan, Hon S. 2004. "Cadre Personnel Management in China: The Nomenklatura System, 1990–1998." *The China Quarterly* 179:703–734. <https://doi.org/10.1017/S0305741004000554>.
- Chan, Stephen. 2003. *Robert Mugabe: A Life of Power and Violence*. University of Michigan Press.
- Chen, Ling, and Hao Zhang. 2021. "Strategic Authoritarianism: The Political Cycles and Selectivity of China's Tax-Break Policy." *American Journal of Political Science* 65, no. 4 (October 1, 2021): 845–861. <https://doi.org/10.1111/ajps.12648>.
- Chen, Wei, Xilu Chen, Chang-Tai Hsieh, and Zheng Song. 2019. *A Forensic Examination of China's National Accounts*. National Bureau of Economic Research. <https://doi.org/10.3386/w25754>.
- Chen, Ye, Hongbin Li, and Li-An Zhou. 2005. "Relative Performance Evaluation and the Turnover of Provincial Leaders in China." *Economics Letters* 88, no. 3 (September 1, 2005): 421–425. <https://doi.org/10.1016/j.econlet.2005.05.003>.
- Chen, Zuoqi, Lingxing Liao, Congxiao Wang, Kaifang Shi, and Bailang Yu. 2026. "The 1992–2024 Global NPP-VIIRS-like Nighttime Light Annual Data from Deep Learning Super-Resolution Reconstruction." *Journal of Remote Sensing* 6 (March 31, 2026): 0874. <https://doi.org/10.34133/remotesensing.0874>.
- Cheng, Rui, Huifang Zhang, and Baozhang Chen. 2025. "China Temporal Administrative Map: A Multitemporal Database for Chinese Historical Administrative Divisions (2009–2023)," vol. 13565, 135651R. *Proc.SPIE*. April 15, 2025. <https://doi.org/10.1117/12.3059430>.
- Chin, John, Abel Escribà-Folch, Wonjun Song, and Joseph Wright. 2022. "Reshaping the Threat Environment: Personalism, Coups, and Assassinations." *Comparative Political Studies* 55, no. 4 (March 1, 2022): 657–687. <https://doi.org/10.1177/00104140211024287>.

- Choi, Eun Kyong, John Wagner Givens, and Andrew MacDonald. 2021. "From Power Balance to Dominant Faction in Xi Jinping's China." *The China Quarterly* 248 (1): 935–956. <https://doi.org/10.1017/S0305741021000473>.
- Dimitrov, Martin K. 2023. *Dictatorship and Information: Authoritarian Regime Resilience in Communist Europe and China*. Oxford University Press.
- Ding, Chengri. 2003. "Land Policy Reform in China: Assessment and Prospects." *Land Use Policy* 20, no. 2 (April 1, 2003): 109–120. [https://doi.org/10.1016/S0264-8377\(02\)00073-X](https://doi.org/10.1016/S0264-8377(02)00073-X).
- Egorov, Georgy, and Konstantin Sonin. 2011. "Dictators and Their Viziers: Endogenizing the Loyalty–Competence Trade-off." *Journal of the European Economic Association* 9, no. 5 (October 1, 2011): 903–930. <https://doi.org/10.1111/j.1542-4774.2011.01033.x>.
- . 2026. "Why Did Putin Invade Ukraine? A Theory of Degenerate Autocracy." *American Journal of Political Science* (February 17, 2026). <https://doi.org/10.1111/ajps.70039>.
- Feng, Bo, Bei Lu, Zhen Wang, and Dandan Yu. 2025. "Patronage Networks and Multitasking Incentives: Evidence from Local Officials' Responses to Public Crises in China's Centralized Bureaucracy." *World Development* 190 (June 1, 2025): 106953. <https://doi.org/10.1016/j.worlddev.2025.106953>.
- Fewsmith, Joseph, and Andrew J. Nathan. 2019. "Authoritarian Resilience Revisited: Joseph Fewsmith with Response from Andrew J. Nathan." *Journal of Contemporary China* 28, no. 116 (March 4, 2019): 167–179. <https://doi.org/10.1080/10670564.2018.1511390>.
- Frantz, Erica, Andrea Kendall-Taylor, Joseph Wright, and Xu Xu. 2020. "Personalization of Power and Repression in Dictatorships." *The Journal of Politics* 82, no. 1 (January 1, 2020): 372–377. <https://doi.org/10.1086/706049>.

- Gandhi, Jennifer. 2008. *Political Institutions under Dictatorship*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511510090>.
- Gandhi, Jennifer, and Jane Lawrence Sumner. 2020. "Measuring the Consolidation of Power in Nondemocracies." *The Journal of Politics* 82, no. 4 (October 1, 2020): 1545–1558. <https://doi.org/10.1086/708683>.
- Geddes, Barbara, Joseph Wright, and Erica Frantz. 2014. "Autocratic Breakdown and Regime Transitions: A New Data Set." *Perspectives on Politics* 12 (2): 313–331. <https://doi.org/10.1017/S1537592714000851>.
- . 2018. *How Dictatorships Work: Power, Personalization, and Collapse*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/9781316336182>.
- Gehlbach, Scott, and Philip Keefer. 2011. "Investment without Democracy: Ruling-party Institutionalization and Credible Commitment in Autocracies." *Journal of Comparative Economics* 39, no. 2 (June 1, 2011): 123–139. <https://doi.org/10.1016/j.jce.2011.04.002>.
- . 2012. "Private Investment and the Institutionalization of Collective Action in Autocracies: Ruling Parties and Legislatures." *The Journal of Politics* 74, no. 2 (April 1, 2012): 621–635. <https://doi.org/10.1017/S0022381611001952>.
- Goldring, Edward, and Austin S. Matthews. 2023. "To Purge or Not to Purge? An Individual-Level Quantitative Analysis of Elite Purges in Dictatorships." *British Journal of Political Science* 53 (2): 575–593. <https://doi.org/10.1017/S0007123421000569>.
- Grundholm, Alexander Taaning. 2020. "Taking It Personal? Investigating Regime Personalization as an Autocratic Survival Strategy." *Democratization* 27, no. 5 (July 3, 2020): 797–815. <https://doi.org/10.1080/13510347.2020.1737677>.

- Guo, Gang. 2009. "China's Local Political Budget Cycles." *American Journal of Political Science* 53 (3): 621–632.
- Henderson, J. Vernon, Adam Storeygard, and David N. Weil. 2012. "Measuring Economic Growth from Outer Space." *American Economic Review* 102 (2): 994–1028. <https://doi.org/10.1257/aer.102.2.994>.
- Hood, Christopher. 2011. *The Blame Game*. Princeton University Press.
- Hu, Yingyao, and Jiaxiong Yao. 2022. "Illuminating Economic Growth." *Journal of Econometrics* 228, no. 2 (June 1, 2022): 359–378. <https://doi.org/10.1016/j.jeconom.2021.05.007>.
- Jackson, Matthew O., and Massimo Morelli. 2007. "Political Bias and War." *American Economic Review* 97 (4): 1353–1373. <https://doi.org/10.1257/aer.97.4.1353>.
- Jia, Junxue, Qingwang Guo, and Jing Zhang. 2014. "Fiscal Decentralization and Local Expenditure Policy in China." *China Economic Review* 28 (March 1, 2014): 107–122. <https://doi.org/10.1016/j.chieco.2014.01.002>.
- Jia, Kai, and Shaowei Chen. 2019. "Could Campaign-Style Enforcement Improve Environmental Performance? Evidence from China's Central Environmental Protection Inspection." *Journal of Environmental Management* 245 (September 1, 2019): 282–290. <https://doi.org/10.1016/j.jenvman.2019.05.114>.
- Jia, Ruixue, Masayuki Kudamatsu, and David Seim. 2015. "Political Selection in China: The Complementary Roles of Connections and Performance." *Journal of the European Economic Association* 13, no. 4 (August 1, 2015): 631–668. <https://doi.org/10.1111/jeea.12124>.
- Jiang, Junyan. 2018. "Making Bureaucracy Work: Patronage Networks, Performance Incentives, and Economic Development in China." *American Journal of Political Science* 62, no. 4 (October 1, 2018): 982–999. <https://doi.org/10.1111/ajps.12394>.

- Jiang, Junyan, and Songpo Yang. 2026. "Portraits of Power: Facial Appearance and the Tacit Domain of Political Selection in China." *American Political Science Review*, 1–22. <https://doi.org/10.1017/S0003055426101749>.
- Jiang, Junyan, and Muyang Zhang. 2020. "Friends with Benefits: Patronage Networks and Distributive Politics in China." *Journal of Public Economics* 184 (April 1, 2020): 104143. <https://doi.org/10.1016/j.jpubeco.2020.104143>.
- Keller, Franziska Barbara. 2016. "Moving Beyond Factions: Using Social Network Analysis to Uncover Patronage Networks Among Chinese Elites." *Journal of East Asian Studies* 16 (1): 17–41. <https://doi.org/10.1017/jea.2015.3>.
- Kendall-Taylor, Andrea, Erica Frantz, and Joseph Wright. 2017. "The Global Rise of Personalized Politics: It's Not Just Dictators Anymore." *The Washington Quarterly* 40, no. 1 (January 2, 2017): 7–19. <https://doi.org/10.1080/0163660X.2017.1302735>.
- Kuehn, David, Mariana Llanos, and Thomas Richter. 2026. "The Personalization of Political Power Revisited." *Democratization* (March 5, 2026): 1–14. <https://doi.org/10.1080/13510347.2026.2636039>.
- Landry, Pierre F. 2008. *Decentralized Authoritarianism in China: The Communist Party's Control of Local Elites in the Post-Mao Era*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511510243>.
- Landry, Pierre F., Xiaobo Lü, and Haiyan Duan. 2018. "Does Performance Matter? Evaluating Political Selection Along the Chinese Administrative Ladder." *Comparative Political Studies* 51, no. 8 (July 1, 2018): 1074–1105. <https://doi.org/10.1177/0010414017730078>.
- Lee, Jonghyuk. 2025. "Limited Power-Sharing in a Personalized Regime: Xi Jinping's Strategic Retreat from Local Politics." *Journal of Contemporary China* 34, no. 152 (March 4, 2025): 253–270. <https://doi.org/10.1080/10670564.2024.2334898>.

- Li, Cheng. 2002. "Hu's Followers: Provincial Leaders with Backgrounds in the Youth League." *China Leadership Monitor* 3 (1): 1–11.
- . 2014a. "Xi Jinping's Inner Circle (Part 1: The Shaanxi Gang)." *China Leadership Monitor* 43, no. 43 (March 14, 2014): 1–21.
- . 2014b. "Xi Jinping's Inner Circle Part 3: Political Protégés from the Provinces." *China Leadership Monitor* 45, no. 45 (October 21, 2014): 1–18.
- Li, Hongbin, and Li-An Zhou. 2005. "Political Turnover and Economic Performance: The Incentive Role of Personnel Control in China." *Journal of Public Economics* 89, no. 9 (September 1, 2005): 1743–1762. <https://doi.org/10.1016/j.jpubeco.2004.06.009>.
- Li, Jenny. 2022. "China Under Xi Jinping." *Journal of International Affairs* 75 (1): 261–272.
- Li, Ling. 2019. "Politics of Anticorruption in China: Paradigm Change of the Party's Disciplinary Regime 2012–2017." *Journal of Contemporary China* 28, no. 115 (January 2, 2019): 47–63. <https://doi.org/10.1080/10670564.2018.1497911>.
- Liu, Zhiwei. 2026. "New Leader, New Policy? Bureaucratic Turnover and Policy Stability in China." *Governance* 39, no. 1 (January 1, 2026): e70094. <https://doi.org/10.1111/gove.70094>.
- Lorentzen, Peter L. 2013. "Regularizing Rioting: Permitting Public Protest in an Authoritarian Regime." *Quarterly Journal of Political Science* 8 (2): 127–158.
- Lyu, Changjiang, Kemin Wang, Frank Zhang, and Xin Zhang. 2018. "GDP Management to Meet or Beat Growth Targets." *Journal of Accounting and Economics* 66, no. 1 (August 1, 2018): 318–338. <https://doi.org/10.1016/j.jacceco.2018.07.001>.
- Mahdavi, Paasha, and John Ishiyama. 2020. "Dynamics of the Inner Elite in Dictatorships." *Comparative Politics* 52 (2): 221–240.

- Malesky, Edmund, and Paul Schuler. 2011. "The Single-Party Dictator's Dilemma: Information in Elections without Opposition." *Legislative Studies Quarterly* 36, no. 4 (November 1, 2011): 491–530. <https://doi.org/10.1111/j.1939-9162.2011.00025.x>.
- Manion, Melanie. 1985. "The Cadre Management System, Post-Mao: The Appointment, Promotion, Transfer and Removal of Party and State Leaders." *The China Quarterly*, no. 102, 203–233.
- . 2023. *Political Selection in China: Rethinking Foundations and Findings*. Elements in Politics and Society in East Asia. Cambridge: Cambridge University Press. <https://doi.org/10.1017/9781009327145>.
- Márquez, Xavier. 2024. "The Mechanisms of Personalization." *Democratization*, 1–21. <https://doi.org/10.1080/13510347.2024.2426197>.
- Martínez, Luis R. 2022. "How Much Should We Trust the Dictator's GDP Growth Estimates?" *Journal of Political Economy* 130, no. 10 (October 1, 2022): 2731–2769. <https://doi.org/10.1086/720458>.
- Meng, Anne. 2020. *Constraining Dictatorship: From Personalized Rule to Institutionalized Regimes*. Political Economy of Institutions and Decisions. Cambridge: Cambridge University Press. <https://doi.org/10.1017/9781108877497>.
- Oliveros, Virginia. 2021. "Working for the Machine: Patronage Jobs and Political Services in Argentina." *Comparative Politics* 53, no. 3 (April 1, 2021): 381–427. <https://doi.org/10.5129/01041521X15974977783469>.
- Ong, Lynette H. 2014. "State-Led Urbanization in China: Skyscrapers, Land Revenue and 'Concentrated Villages'." *The China Quarterly* 217:162–179. <https://doi.org/10.1017/S0305741014000010>.

- Paine, Jack. 2021. "The Dictator's Power-Sharing Dilemma: Countering Dual Outsider Threats." *American Journal of Political Science* 65, no. 2 (April 1, 2021): 510–527. <https://doi.org/10.1111/ajps.12547>.
- Que, Wei, Yabin Zhang, and Günther Schulze. 2019. "Is Public Spending Behavior Important for Chinese Official Promotion? Evidence from City-Level." *China Economic Review* 54 (April 1, 2019): 403–417. <https://doi.org/10.1016/j.chieco.2019.02.003>.
- Reuter, Ora John, Noah Buckley, Alexandra Shubenkova, and Guzel Garifullina. 2016. "Local Elections in Authoritarian Regimes: An Elite-Based Theory With Evidence From Russian Mayoral Elections." *Comparative Political Studies* 49, no. 5 (April 1, 2016): 662–697. <https://doi.org/10.1177/0010414015626439>.
- Reuter, Ora John, and Graeme B. Robertson. 2012. "Subnational Appointments in Authoritarian Regimes: Evidence from Russian Gubernatorial Appointments." *The Journal of Politics* 74, no. 4 (October 1, 2012): 1023–1037. <https://doi.org/10.1017/S0022381612000631>.
- Scott, James C. 1972. "Patron-Client Politics and Political Change in Southeast Asia." *The American Political Science Review* 66 (1): 91–113. <https://doi.org/10.2307/1959280>.
- Shen, Yang, Jing Wu, and Shuping Wu. 2022. "City-Chief Turnover and Place-Based Policy Change: Evidence from China." *Journal of Regional Science* 62, no. 5 (November 1, 2022): 1296–1328. <https://doi.org/10.1111/jors.12598>.
- Shih, Victor, Christopher Adolph, and Mingxing Liu. 2012. "Getting Ahead in the Communist Party: Explaining the Advancement of Central Committee Members in China." *American Political Science Review* 106 (1): 166–187. <https://doi.org/10.1017/S0003055411000566>.
- Shirk, Susan L. 2018. "China in Xi's 'New Era': The Return to Personalistic Rule." *Journal of Democracy* 29 (2): 22–36.

- Shu, Cheng, Hualin Xie, Jinfa Jiang, and Qianru Chen. 2018. "Is Urban Land Development Driven by Economic Development or Fiscal Revenue Stimuli in China?" *Land Use Policy* 77 (September 1, 2018): 107–115. <https://doi.org/10.1016/j.landusepol.2018.05.031>.
- Smith, Graeme. 2018. "The Campaign Rolls on: Rural Governance in China under Xi Jinping and the War on Poverty." *China: An International Journal* 16 (3): 163–178.
- Sudduth, Jun Koga. 2017. "Strategic Logic of Elite Purges in Dictatorships." *Comparative Political Studies* 50, no. 13 (November 1, 2017): 1768–1801. <https://doi.org/10.1177/0010414016688004>.
- Svolik, Milan W. 2009. "Power Sharing and Leadership Dynamics in Authoritarian Regimes." *American Journal of Political Science* 53 (2): 477–494.
- . 2012. *The Politics of Authoritarian Rule*. Cambridge Studies in Comparative Politics. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781139176040>.
- Timoneda, Joan C., Abel Escribà-Folch, and Kyle Haynes. 2025. "The Trajectories of Personalization: How Dictators Accumulate Power over Time." *Democratization* (November 7, 2025): 1–24. <https://doi.org/10.1080/13510347.2025.2581837>.
- Tullock, Gordon. 1987. *Autocracy*. Springer Science & Business Media.
- Wallace, Jeremy L. 2016. "Juking the Stats? Authoritarian Information Problems in China." *British Journal of Political Science* 46 (1): 11–29. <https://doi.org/10.1017/S0007123414000106>.
- Wang, Erik H. 2022. "Frightened Mandarins: The Adverse Effects of Fighting Corruption on Local Bureaucracy." *Comparative Political Studies* 55, no. 11 (September 1, 2022): 1807–1843. <https://doi.org/10.1177/00104140211060276>.
- Weaver, R. Kent. 1986. "The Politics of Blame Avoidance." *Journal of Public Policy* 6 (4): 371–398.

- Weeks, Jessica L. 2012. “Strongmen and Straw Men: Authoritarian Regimes and the Initiation of International Conflict.” *The American Political Science Review* 106 (2): 326–347.
- Wintrobe, Ronald. 1998. *The Political Economy of Dictatorship*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781139174916>.
- Wright, Joseph. 2008. “Do Authoritarian Institutions Constrain? How Legislatures Affect Economic Growth and Investment.” *American Journal of Political Science* 52, no. 2 (April 1, 2008): 322–343. <https://doi.org/10.1111/j.1540-5907.2008.00315.x>.
- Wu, Qun, Yongle Li, and Siqi Yan. 2015. “The Incentives of China’s Urban Land Finance.” *Land Use Policy* 42 (January 1, 2015): 432–442. <https://doi.org/10.1016/j.landusepol.2014.08.015>.
- Xu, Chenggang. 2011. “The Fundamental Institutions of China’s Reforms and Development.” *Journal of Economic Literature* 49 (4): 1076–1151. <https://doi.org/10.1257/jel.49.4.1076>.
- Yao, Yang, and Muyang Zhang. 2015. “Subnational Leaders and Economic Growth: Evidence from Chinese Cities.” *Journal of Economic Growth* 20, no. 4 (December 1, 2015): 405–436. <https://doi.org/10.1007/s10887-015-9116-1>.
- Zakharov, Alexei V. 2016. “The Loyalty-Competence Trade-Off in Dictatorships and Outside Options for Subordinates.” *The Journal of Politics* 78, no. 2 (April 1, 2016): 457–466. <https://doi.org/10.1086/684365>.
- Zeng, Jiangnan, and Qiyao Zhou. 2024. “Mayors’ Promotion Incentives and Subnational-Level GDP Manipulation.” *Journal of Urban Economics* 143 (September 1, 2024): 103679. <https://doi.org/10.1016/j.jue.2024.103679>.
- Zeng, Qingjie. 2016. “Democratic Procedures in the CCP’s Cadre Selection Process: Implementation and Consequences.” *The China Quarterly* 225:73–99. <https://doi.org/10.1017/S0305741015001587>.

Zhang, Siyi, Lingna Zhong, Shu Keng, and Baoqing Pang. 2025. "Parsing Factional Affiliations among Chinese Political Elites." *The China Quarterly* 263:595–615. <https://doi.org/10.1017/S0305741025100957>.

Supplemental Materials

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Appendix A: Political Connection Coding

Table A.1: Xi Protégé Provincial Party Secretaries

Name	Name in English	Province	Tenure Start	Tenure End
巴音朝鲁	Bayanqolu	Jilin	August 2014	November 2020
黄兴国	Huang Xingguo	Tianjin	December 2014	September 2016
陈敏尔	Chen Miner	Guizhou	July 2015	July 2017
李希	Li Xi	Liaoning	May 2015	October 2017
杜家毫	Du Jiahao	Hunan	August 2016	November 2020
李强	Li Qiang	Jiangsu	June 2016	October 2017
陈豪	Chen Hao	Yunnan	August 2016	November 2020
蔡奇	Cai Qi	Beijing	May 2017	November 2022
陈敏尔	Chen Miner	Chongqing	July 2017	December 2022
于伟国	Yu Weiguo	Fujian	October 2017	November 2020
李希	Li Xi	Guangdong	October 2017	October 2022
刘赐贵	Liu Cigui	Hainan	April 2017	November 2020
石泰峰	Shi Taifeng	Ningxia Hui	April 2017	October 2019
李强	Li Qiang	Shanghai	October 2017	October 2022
刘奇	Liu Qi	Jiangxi	March 2018	October 2021
石泰峰	Shi Taifeng	Nei Mongol	October 2019	April 2022
楼阳生	Lou Yangsheng	Shanxi	November 2019	June 2021
应勇	Ying Yong	Hubei	February 2020	March 2022
郑栅洁	Zheng Shanjie	Anhui	September 2021	March 2023
尹弘	Yin Hong	Gansu	March 2021	December 2022
楼阳生	Lou Yangsheng	Henan	June 2021	December 2024
黄坤明	Huang Kunming	Guangdong	October 2022	December 2024*
徐麟	Xu Lin	Guizhou	December 2022	December 2024*
尹弘	Yin Hong	Jiangxi	December 2022	December 2024*
梁言顺	Liang Yanshun	Ningxia Hui	March 2022	June 2024
赵一德	Zhao Yide	Shaanxi	November 2022	December 2024*
陈敏尔	Chen Miner	Tianjin	October 2022	December 2024*
沈晓明	Shen Xiaoming	Hunan	March 2023	December 2024*

Notes: Asterisks indicate right-censored provincial party-secretary tenures that were still ongoing at the end of the study window in December 2024. Huang Xingguo was acting Party secretary of Tianjin throughout the listed tenure.

Table A.2: Hu-Connected Provincial Party Secretaries

Name	Name in English	Province	Tenure Start	Tenure End
李克强	Li Keqiang	Liaoning	December 2004	October 2007
杨传堂	Yang Chuantang	Tibet	December 2004	May 2006
汪洋	Wang Yang	Chongqing	December 2005	November 2007
张宝顺	Zhang Baoshun	Shanxi	July 2005	May 2010
张庆黎	Zhang Qingli	Tibet	November 2005	August 2011
刘奇葆	Liu Qibao	Guangxi	June 2006	December 2007
韩正	Han Zheng	Shanghai	September 2006	March 2007
杜青林	Du Qinglin	Sichuan	December 2006	December 2007
汪洋	Wang Yang	Guangdong	December 2007	December 2012
强卫	Qiang Wei	Qinghai	March 2007	March 2013
刘奇葆	Liu Qibao	Sichuan	December 2007	November 2012
吉炳轩	Ji Bingxuan	Heilongjiang	April 2008	March 2013
胡春华	Hu Chunhua	Nei Mongol	November 2009	December 2012
张宝顺	Zhang Baoshun	Anhui	May 2010	June 2015
栗战书	Li Zhanshu	Guizhou	August 2010	July 2012
周强	Zhou Qiang	Hunan	April 2010	March 2013
罗志军	Luo Zhijun	Jiangsu	December 2010	June 2016
袁纯清	Yuan Chunqing	Shanxi	May 2010	September 2014
王三运	Wang Sanyun	Gansu	December 2011	April 2017
罗保铭	Luo Baoming	Hainan	August 2011	April 2017
张庆黎	Zhang Qingli	Hebei	August 2011	March 2013
秦光荣	Qin Guangrong	Yunnan	August 2011	October 2014
胡春华	Hu Chunhua	Guangdong	December 2012	October 2017
王儒林	Wang Rulin	Jilin	December 2012	August 2014
韩正	Han Zheng	Shanghai	November 2012	October 2017
强卫	Qiang Wei	Jiangxi	March 2013	June 2016

Notes: Han Zheng was acting Party secretary of Shanghai during the 2006–2007 tenure. Zhang Qingli was acting Party secretary of Tibet from November 2005 to May 2006, accounting for the Yang Chuantang overlap.

Appendix B: Summary Statistics for City-Year Economic Data

Table B.1: Summary Statistics for City-Year Economic Variables

Variable	N	Mean	SD	P1	Median	P99
GDP per capita	5,624	42.41	48.56	3.48	28.37	246.14
Real GDP per capita growth	5,001	0.09	0.08	-0.21	0.09	0.28
Population	5,678	4159.78	2489.66	429.08	3644.20	11479.46
Year-end outstanding loans	6,284	269.39	615.79	6.45	89.03	3057.02
Real year-end loan growth	5,990	0.11	0.15	-0.15	0.11	0.41
Nighttime lights, boundary sum	7,630	72.74	64.44	2.33	53.01	300.18
Nighttime-light growth	7,295	0.07	0.34	-1.28	0.06	1.00
Total electricity consumption	5,080	11.51	16.04	0.25	5.64	79.58
Electricity consumption growth	4,714	0.13	0.37	-0.59	0.08	1.77
Fiscal expenditure	6,557	28.82	35.80	1.12	19.58	171.94
Fiscal expenditure growth	6,261	0.13	0.12	-0.17	0.13	0.42
Land-transfer revenue	7,389	21.32	561.46	0.00	2.74	153.97
Land-revenue expansion	6,920	0.58	0.49	0.00	1.00	1.00

Notes: The table reports descriptive statistics for city-year variables used to construct the leader-level performance profiles and the pre-selection outcomes, before outcome-specific trimming, connected-component restrictions, and conversion to leader-year-role observations. For each variable, unobserved city-year values are coded as missing and excluded from the corresponding descriptive statistics. Monetary variables are reported in billion yuan, GDP per capita in thousand yuan per person, population in thousand persons, electricity consumption in billion kWh, and nighttime-light sums in thousand luminosity units. Growth variables are annual log changes. Land-revenue expansion equals one when observed city-level land-transfer revenue exceeds its value in the immediately preceding calendar year and zero otherwise; it is coded as missing when revenue is unavailable in either year. P1 and P99 report the 1st and 99th percentiles, respectively. Zero population values are treated as missing in the descriptive summary.

Appendix C: Revealed Performance Profile Estimation

Table C.1: Fixed-Effect Models Used to Estimate Revealed Performance Profiles

Model	(1)	(2)	(3)	(4)
Dependent variable	Real GDP per capita growth	Real year-end loan growth	Boundary nighttime-light growth	GDP–NTL residual
Generated leader profile	Real GDP per capita growth profile	Real year-end loan growth profile	Nighttime-light growth profile	GDP–NTL residual profile
Initial log level	-0.017***	-0.018***	-0.058***	-0.017***
(S.E.)	(0.005)	(0.004)	(0.012)	(0.005)
Log population	-0.195***	-0.075	0.052	-0.196***
(S.E.)	(0.049)	(0.044)	(0.069)	(0.049)
Leader fixed effects	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes
Province-year fixed effects	Yes	Yes	Yes	Yes
Clustered standard errors	City	City	City	City
Observations	8,999	9,501	10,074	8,999
Unique leaders	1,842	2,080	2,168	1,842
Cities	283	288	288	283
Years	2003–2022	2004–2022	2003–2022	2003–2022
R-squared	0.767	0.756	0.801	0.600
Adjusted R-squared	0.674	0.653	0.718	0.441

Notes: The dependent variable in each column is the city-year outcome used to estimate the corresponding leader fixed-effect profile. Entries report coefficient estimates with city-clustered standard errors in parentheses below estimates. Observations are leader-year-role observations because each city-year outcome is linked separately to the mayor and the prefectural party secretary. The generated profile is the centered leader fixed effect multiplied by 100. As a diagnostic, each full model is compared with a restricted model that omits leader fixed effects. The F-tests reject the restricted model for real GDP per capita growth, real year-end loan growth, and the GDP–NTL residual outcome, with $F = 1.55$, $F = 1.44$, and $F = 1.54$, respectively, and $p < 0.001$ in all three cases. The same diagnostic does not reject the restricted model for boundary nighttime-light growth, with $F = 0.46$.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table C.2: First-Stage Residualization of Reported GDP per Capita Growth on Nighttime-Light Growth

Model	(1)
Dependent variable	Real GDP per capita growth
Boundary nighttime-light growth (S.E.)	0.004 (0.004)
City fixed effects	Yes
Year fixed effects	Yes
Clustered standard errors	City
City-year observations	4,874
Cities	289
Years	2003–2022
R-squared	0.415
Adjusted R-squared	0.376

Notes: The dependent variable is real GDP per capita growth. The residual from this city-year model is used to estimate the leader-level GDP–NTL residual profile. Standard errors are clustered by city.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table C.3: Diagnostic Models for the GDP–Nighttime-Light Association

Model	Dependent variable: Real GDP per capita growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Boundary nighttime-light growth (S.E.)	0.021*** (0.003)	0.008* (0.004)	0.020*** (0.003)	0.004 (0.004)	0.004 (0.005)	0.002 (0.005)
Year fixed effects	No	Yes	No	Yes	No	No
City fixed effects	No	No	Yes	Yes	No	Yes
Province-year fixed effects	No	No	No	No	Yes	Yes
Clustered standard errors	City	City	City	City	City	City
City-year observations	4,874	4,874	4,874	4,874	4,848	4,848
R-squared	0.009	0.351	0.074	0.415	0.630	0.658
Within R-squared	—	0.001	0.009	0.000	0.000	0.000

Notes: The table reports diagnostic regressions of reported real GDP per capita growth on boundary nighttime-light growth under alternative fixed-effect structures. Standard errors clustered by city are shown in parentheses. Models with province-year fixed effects drop 26 singleton observations.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix D: Estimated Leader-Level Revealed Performance Profiles and Match Coverage

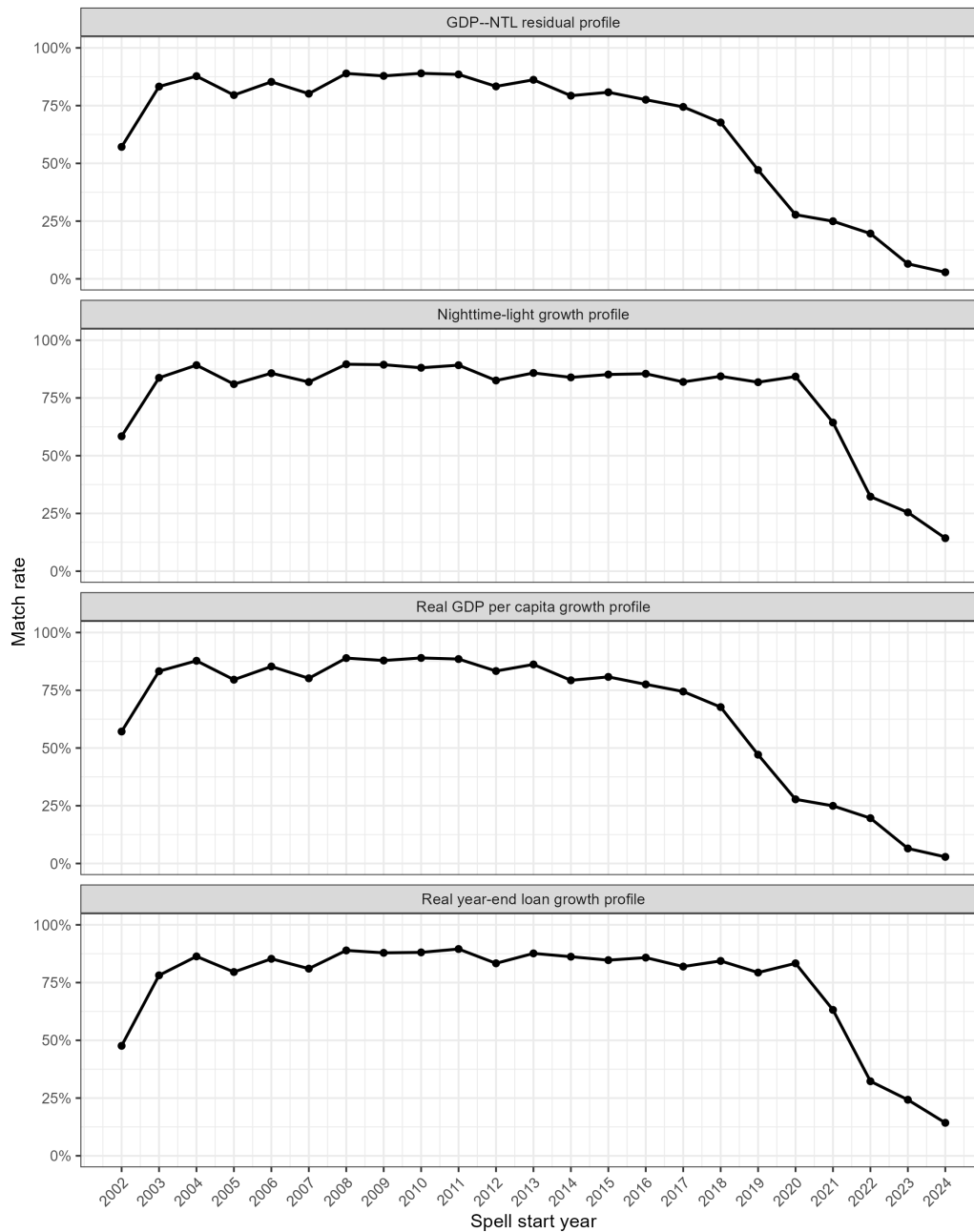


Figure D.1: Performance-Profile Match Rates by Leader-Spell Start Year

Notes: The figure reports the share of mayoral and prefectural party-secretary spells that can be matched to each leader-level revealed performance profile, by spell start year. Match rates decline for later cohorts, especially for GDP-based profiles after 2018 and across profiles after 2019, reflecting limited usable outcome histories for spells that began recently. This pattern motivates restricting the main selection analyses to first appointments made no later than 2019.

Table D.1: Distribution of Leader-Level Revealed Performance Profiles

Profile	N leaders	Mean	SD	Min	Median	Max
Real GDP per capita growth profile	1,842	0.00	4.99	-28.59	0.13	18.15
Real year-end loan growth profile	2,080	0.00	5.91	-26.91	0.49	21.15
Nighttime-light growth profile	2,168	0.00	9.24	-34.11	-0.39	73.65
GDP–NTL residual profile	1,842	0.00	5.01	-28.52	0.10	18.35

Notes: Profiles are centered leader fixed effects multiplied by 100, so units are approximately percentage points. Zero corresponds to the mean leader fixed effect within the connected component of each estimation sample. Higher GDP–NTL residual values indicate greater reported real GDP per capita growth relative to the nighttime-light benchmark.

Appendix E: Selection Sample and Descriptive Comparisons

Table E.1: Selection Sample Composition and Leader Characteristics by Era and Promoter Type

Statistic	Hu-connected PPS (2004–2013)	Non-connected PPS (2004–2013)	Xi-protégé PPS (2014–2019)	Non-protégé PPS (2014–2019)
Promoted city leaders	416	1,056	157	780
Unique individuals	378	887	146	700
Provinces	19	27	11	27
Cities	190	291	84	286
Provincial party secretaries	17	55	10	44
Provincial party-secretary tenures	22	62	11	53
Mayor promotions	220	588	85	413
Prefectural party-secretary promotions	196	468	72	367
Age at appointment	48.9	48.4	50.9	51.1
Male (%)	92.5	93.7	94.3	94.1
Master/graduate degree (%)	56.6	56.5	56.7	60.5
Doctorate (%)	23.1	20.2	26.1	22.9

Notes: The table reports sample composition and descriptive characteristics for role-specific first appointments as mayor or prefectural party secretary. Appointments are attributed to provincial party secretaries using the three-month buffer rule. Columns correspond to the main selection-analysis windows: 2004–2013 for the Hu-era benchmark and 2014–2019 for the Xi-era analysis. Unique individuals are counted by leader identifier. Age is reported as a mean among observations with nonmissing age information. Gender and education entries are percentages calculated separately among observations with nonmissing information for the corresponding characteristic. PPS denotes provincial party secretary.

Table E.2: Revealed Performance Profiles by Era and Promoter Type

Statistic	Hu-connected PPS (2004–2013)	Non-connected PPS (2004–2013)	Xi-protégé PPS (2014–2019)	Non-protégé PPS (2014–2019)
GDP–NTL residual profile	0.76 (348)	-0.00 (910)	-0.79 (91)	1.11 (567)
Real GDP per capita growth profile	0.77 (348)	-0.01 (910)	-0.76 (91)	1.15 (567)
Real year-end loan growth profile	-0.60 (351)	-0.10 (909)	1.78 (124)	0.36 (648)
Nighttime-light growth profile	1.01 (345)	-0.35 (918)	1.58 (125)	0.09 (647)

Notes: Entries report group means of leader-level revealed performance profiles; matched spell counts are shown in parentheses. Profiles are centered leader fixed effects multiplied by 100, so units are approximately percentage points. Higher GDP–NTL residual values indicate greater reported real GDP per capita growth relative to the nighttime-light benchmark. PPS denotes provincial party secretary.

Table E.3: Match-Coverage Models for Leader-Level Performance Profiles

Model	(1)	(2)	(3)	(4)
Dependent variable	Matched to real GDP per capita growth profile	Matched to real year-end loan growth profile	Matched to nighttime-light growth profile	Matched to GDP-NTL residual profile
Leader profile	Real GDP per capita growth profile	Real year-end loan growth profile	Nighttime-light growth profile	GDP-NTL residual profile
Xi-protégé PPS (S.E.)	-0.009 (0.043)	-0.076* (0.035)	-0.062* (0.030)	-0.009 (0.043)
Province fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Role fixed effects	Yes	Yes	Yes	Yes
Clustered standard errors	Province	Province	Province	Province
Observations	937	937	937	937
Provinces	27	27	27	27
Years	2014–2019	2014–2019	2014–2019	2014–2019
Mean dependent variable	0.702	0.824	0.824	0.702

Notes: The table reports linear probability models for the Xi-era selection sample. In each column, the dependent variable equals one if a first-time city-leader promotion can be matched to the corresponding leader-level performance profile. The coefficient on Xi-protégé PPS indicates whether promotions under Xi-protégé provincial party secretaries are more or less likely to have a matched profile than promotions under non-protégé provincial party secretaries, conditional on province, year, and role fixed effects. Standard errors are clustered by province. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix F: Selection Model Results

Table F.1: Xi-Era Selection Models across Revealed Performance Profiles

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable	GDP growth	Loan growth	NTL growth	GDP–NTL residual	GDP growth	Loan growth	NTL growth	GDP–NTL residual
3-month buffer	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Xi-protégé PPS	-1.076*** (0.264) [0.004]	0.703 (0.547) [0.232]	-0.088 (0.692) [0.903]	-1.096*** (0.268) [0.004]	-1.104*** (0.283) [0.004]	0.687 (0.586) [0.286]	-0.160 (0.753) [0.843]	-1.123*** (0.286) [0.004]
Male					1.710* (0.745)	1.303* (0.583)	0.178 (0.900)	1.707* (0.748)
Master/graduate degree					-0.058 (0.530)	0.208 (0.775)	-1.003 (0.901)	-0.057 (0.534)
Doctorate					-0.014 (0.702)	0.371 (0.901)	-0.363 (1.130)	-0.005 (0.703)
Age at appointment					-0.031 (0.074)	-0.004 (0.071)	0.090 (0.096)	-0.029 (0.074)
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Role fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered standard errors	Province	Province	Province	Province	Province	Province	Province	Province
Observations	658	772	772	658	658	772	772	658
Unique leaders	564	665	665	564	564	665	665	564
Provinces	25	26	26	25	25	26	26	25
Cities	254	262	262	254	254	262	262	254
Mayor promotions	332	406	406	332	332	406	406	332
Prefectural party-secretary promotions	326	366	366	326	326	366	366	326
Promotions by Xi-protégé PPS	91	124	125	91	91	124	125	91
Promotions by non-protégé PPS	567	648	647	567	567	648	647	567

Notes: The table reports Xi-era selection models across the four leader-level performance profiles using the three-month buffer rule. Columns (1)–(4) report the baseline specifications for real GDP per capita growth, real year-end loan growth, nighttime-light growth, and the GDP–NTL residual, respectively; columns (5)–(8) repeat the same outcomes after adding individual controls for gender, education, and age at appointment. Standard errors clustered by province are shown in parentheses. Wild cluster bootstrap p-values are shown in square brackets for the Xi-protégé PPS coefficient. All models include province, year, and role fixed effects. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table F.2: Hu-Era Selection Models across Revealed Performance Profiles

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable	GDP growth	Loan growth	NTL growth	GDP–NTL residual	GDP growth	Loan growth	NTL growth	GDP–NTL residual
3-month buffer	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hu-connected PPS	0.279 (0.261) [0.326]	-0.141 (0.219) [0.509]	0.416 (0.824) [0.638]	0.278 (0.260) [0.326]	0.277 (0.248) [0.307]	-0.142 (0.224) [0.515]	0.411 (0.824) [0.642]	0.276 (0.248) [0.307]
Male					-0.443 (0.372)	0.153 (0.543)	0.170 (0.600)	-0.454 (0.373)
Master/graduate degree					0.526 (0.394)	0.220 (0.449)	-0.949 (0.633)	0.532 (0.395)
Doctorate					0.459 (0.343)	-0.173 (0.477)	-0.536 (0.894)	0.470 (0.344)
Age at appointment					-0.014 (0.045)	-0.052 (0.046)	-0.048 (0.069)	-0.012 (0.045)
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Role fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered standard errors	Province	Province	Province	Province	Province	Province	Province	Province
Observations	1,258	1,260	1,263	1,258	1,258	1,260	1,263	1,258
Unique leaders	964	965	966	964	964	965	966	964
Provinces	25	26	26	25	25	26	26	25
Cities	265	271	271	265	265	271	271	265
Mayor promotions	691	693	693	691	691	693	693	691
Prefectural party-secretary promotions	567	567	570	567	567	567	570	567
Promotions by Hu-connected PPS	348	351	345	348	348	351	345	348
Promotions by non-connected PPS	910	909	918	910	910	909	918	910

Notes: The table reports Hu-era selection models across the four leader-level performance profiles using the three-month buffer rule. Columns (1)–(4) report the baseline specifications for real GDP per capita growth, real year-end loan growth, nighttime-light growth, and the GDP–NTL residual, respectively; columns (5)–(8) repeat the same outcomes after adding individual controls for gender, education, and age at appointment. Standard errors clustered by province are shown in parentheses. Wild cluster bootstrap p-values are shown in square brackets for the Hu-connected PPS coefficient. All models include province, year, and role fixed effects. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table F.3: Pooled Hu–Xi Selection Models across Revealed Performance Profiles, Three-Month Buffer

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable	GDP growth	Loan growth	NTL growth	GDP–NTL residual	GDP growth	Loan growth	NTL growth	GDP–NTL residual
3-month buffer	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Xi era	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed
Connected/protégé PPS	0.334 (0.271)	0.145 (0.222)	0.293 (0.859)	0.335 (0.271)	0.336 (0.267)	0.139 (0.218)	0.294 (0.867)	0.337 (0.266)
Connected/protégé PPS × Xi era	-1.437*** (0.274) [<0.001]	0.057 (0.577) [0.925]	-0.113 (0.996) [0.915]	-1.448*** (0.278) [<0.001]	-1.432*** (0.278) [<0.001]	0.047 (0.571) [0.936]	-0.198 (0.984) [0.849]	-1.442*** (0.282) [<0.001]
Male					0.189 (0.430)	0.586 (0.404)	0.465 (0.560)	0.181 (0.432)
Master/graduate degree					0.333 (0.336)	0.146 (0.444)	-1.125 (0.588)	0.338 (0.337)
Doctorate					0.341 (0.347)	-0.008 (0.465)	-0.592 (0.830)	0.351 (0.347)
Age at appointment					-0.015 (0.042)	-0.026 (0.047)	-0.028 (0.059)	-0.013 (0.042)
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Role fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered standard errors	Province	Province	Province	Province	Province	Province	Province	Province
Observations	1,916	2,032	2,035	1,916	1,916	2,032	2,035	1,916
Unique leaders	1,382	1,482	1,483	1,382	1,382	1,482	1,483	1,382
Provinces	25	26	26	25	25	26	26	25
Cities	271	276	276	271	271	276	276	271
Mayor promotions	1,023	1,099	1,099	1,023	1,023	1,099	1,099	1,023
Prefectural party-secretary promotions	893	933	936	893	893	933	936	893
Promotions by Hu-connected PPS	348	351	345	348	348	351	345	348
Promotions by non-connected PPS	910	909	918	910	910	909	918	910
Promotions by Xi-protégé PPS	91	124	125	91	91	124	125	91
Promotions by non-protégé PPS	567	648	647	567	567	648	647	567

Notes: The table reports pooled Hu–Xi selection models across the four leader-level performance profiles using the three-month buffer rule. Columns (1)–(4) report the baseline specifications for real GDP per capita growth, real year-end loan growth, nighttime-light growth, and the GDP–NTL residual, respectively; columns (5)–(8) repeat the same outcomes after adding individual controls for gender, education, and age at appointment. Standard errors clustered by province are shown in parentheses. Wild cluster bootstrap p-values are shown in square brackets for the Connected/protégé PPS × Xi era coefficient. All models include province, year, and role fixed effects; the Xi-era indicator is absorbed by year fixed effects. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table F.4: Pooled Hu–Xi Selection Models across Revealed Performance Profiles, No Buffer

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable	GDP growth	Loan growth	NTL growth	GDP–NTL residual	GDP growth	Loan growth	NTL growth	GDP–NTL residual
3-month buffer	No	No	No	No	No	No	No	No
Xi era	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed	Absorbed
Connected/protégé PPS	0.412 (0.253)	0.083 (0.256)	0.202 (0.923)	0.414 (0.252)	0.415 (0.248)	0.081 (0.253)	0.210 (0.932)	0.417 (0.247)
Connected/protégé PPS × Xi era	-1.465*** (0.283) [0.002]	-0.305 (0.620) [0.683]	-0.199 (1.000) [0.853]	-1.487*** (0.286) [0.001]	-1.470*** (0.286) [0.002]	-0.324 (0.616) [0.652]	-0.280 (0.994) [0.794]	-1.491*** (0.289) [0.001]
Male					0.195 (0.429)	0.594 (0.406)	0.470 (0.560)	0.187 (0.430)
Master/graduate degree					0.341 (0.334)	0.143 (0.444)	-1.127 (0.587)	0.346 (0.335)
Doctorate					0.350 (0.346)	-0.008 (0.464)	-0.591 (0.830)	0.360 (0.346)
Age at appointment					-0.014 (0.042)	-0.026 (0.047)	-0.028 (0.059)	-0.013 (0.042)
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Role fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered standard errors	Province	Province	Province	Province	Province	Province	Province	Province
Observations	1,916	2,032	2,035	1,916	1,916	2,032	2,035	1,916
Unique leaders	1,382	1,482	1,483	1,382	1,382	1,482	1,483	1,382
Provinces	25	26	26	25	25	26	26	25
Cities	271	276	276	271	271	276	276	271
Mayor promotions	1,023	1,099	1,099	1,023	1,023	1,099	1,099	1,023
Prefectural party-secretary promotions	893	933	936	893	893	933	936	893
Promotions by Hu-connected PPS	336	339	333	336	336	339	333	336
Promotions by non-connected PPS	922	921	930	922	922	921	930	922
Promotions by Xi-protégé PPS	97	136	136	97	97	136	136	97
Promotions by non-protégé PPS	561	636	636	561	561	636	636	561

Notes: The table reports pooled Hu–Xi selection models across the four leader-level performance profiles using the no-buffer rule. Columns (1)–(4) report the baseline specifications for real GDP per capita growth, real year-end loan growth, nighttime-light growth, and the GDP–NTL residual, respectively; columns (5)–(8) repeat the same outcomes after adding individual controls for gender, education, and age at appointment. Standard errors clustered by province are shown in parentheses. Wild cluster bootstrap p-values are shown in square brackets for the Connected/protégé PPS × Xi era coefficient. All models include province, year, and role fixed effects; the Xi-era indicator is absorbed by year fixed effects. PPS denotes provincial party secretary.

* p < 0.05, ** p < 0.01, *** p < 0.001.

Table F.5: Common-Sample Xi-Era Selection Models across Performance Profiles

Model	(1)	(2)	(3)	(4)
Dependent variable	Real GDP per capita growth profile	Real year-end loan growth profile	Nighttime-light growth profile	GDP-NTL residual profile
Xi-protégé PPS	-1.061*** (0.257) [0.004]	0.415 (0.629) [0.570]	-0.52 (0.709) [0.503]	-1.081*** (0.26) [0.004]
Province fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Role fixed effects	Yes	Yes	Yes	Yes
Clustered standard errors	Province	Province	Province	Province
Observations	656	656	656	656
Unique leaders	562	562	562	562
Provinces	25	25	25	25
Cities	254	254	254	254
Mayor promotions	330	330	330	330
Prefectural party-secretary promotions	326	326	326	326
Promotions by Xi-protégé PPS	91	91	91	91
Promotions by non-protégé PPS	565	565	565	565

Notes: The table reports Xi-era selection models estimated on the common sample of first-time city-leader promotions that can be matched to all four leader-level performance profiles. Standard errors clustered by province are shown in parentheses. Wild cluster bootstrap p-values for the Xi-protégé PPS coefficient are shown in square brackets. Profiles are centered leader fixed effects multiplied by 100, so units are approximately percentage points. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix G: Role Heterogeneity

Table G.1: Role-Specific Pooled Benchmark Models for the GDP–NTL Residual Profile

Model	Dependent variable: GDP–NTL residual profile	
	(1)	(2)
	Prefectural party-secretary promotions	Mayor promotions
Sample		
Connected/protégé PPS	0.474 (0.379)	0.223 (0.297)
Xi era	Absorbed	Absorbed
Connected/protégé PPS $\times$ Xi era	-2.006*** (0.427) [<0.001]	-1.014 (0.609) [0.144]
Province fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Role fixed effects	No	No
Clustered standard errors	Province	Province
Observations	893	1,023
Unique leaders	893	1,023
Provinces	25	25
Cities	264	259
Mayor promotions	—	1,023
Prefectural party-secretary promotions	893	—
Promotions by Hu-connected PPS	163	185
Promotions by non-connected PPS	404	506
Promotions by Xi-protégé PPS	44	47
Promotions by non-protégé PPS	282	285

Notes: The table reports role-specific pooled Hu–Xi benchmark models for the GDP–NTL residual profile, estimated separately for prefectural party-secretary and mayoral promotions. Standard errors clustered by province are shown in parentheses. Wild cluster bootstrap p-values are shown in square brackets for the interaction coefficient. The interaction term captures the additional Xi-era protégé gap relative to the Hu-era connected benchmark within each role-specific sample. The Xi-era indicator is absorbed by year fixed effects. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table G.2: Triple-Interaction Test of Role Heterogeneity for the GDP–NTL Residual Profile

Model	(1)
Dependent variable	GDP–NTL residual profile
Sample	Hu and Xi eras
Model	Triple-interaction role heterogeneity
<i>Panel A: Model estimates</i>	
Connected/protégé PPS	0.223 (0.400)
Connected/protégé PPS $\times$ Xi era	-1.014 (0.639)
Connected/protégé PPS $\times$ Prefectural party-secretary promotion	0.252 (0.552)
Connected/protégé PPS $\times$ Xi era $\times$ Prefectural party-secretary promotion	-0.993 (1.033)
<i>Panel B: Interpretable benchmark contrasts</i>	
Mayor promotions: additional Xi-era protégé gap	-1.014 (0.639)
Prefectural party-secretary promotions: additional Xi-era protégé gap	-2.006* (0.657)
Prefectural party-secretary minus mayor difference	-0.993 (1.033)
Province-by-role fixed effects	Yes
Year-by-role fixed effects	Yes
Clustered standard errors	Province
Inference	CR2 / Satterthwaite / HTZ
Observations	1,916
Unique leaders	1,382
Provinces	25
Cities	271
Mayor promotions	1,023
Prefectural party-secretary promotions	893
Promotions by Hu-connected PPS	348
Promotions by non-connected PPS	910
Promotions by Xi-protégé PPS	91
Promotions by non-protégé PPS	567

Notes: The table reports the triple-interaction role-heterogeneity model for the GDP–NTL residual profile. Panel A reports all estimable coefficients involving the Connected/protégé PPS indicator. Mayor promotions and the Hu era are the omitted role and era categories, respectively. Connected/protégé PPS denotes a Hu-connected provincial party secretary in the Hu era and a Xi-protégé provincial party secretary in the Xi era; promotions are attributed using the three-month buffer rule. The main effects for the Xi era and prefectural party-secretary promotions, as well as their interaction, are absorbed by the fixed-effect structure. Panel B reports interpretable benchmark contrasts: the additional Xi-era protégé gap among mayor promotions, the additional Xi-era protégé gap among prefectural party-secretary promotions, and the prefectural party-secretary-minus-mayor difference. Entries use CR2 small-sample cluster-robust standard errors in parentheses. Stars are based on CR2/Satterthwaite p-values for single coefficients and CR2/HTZ Wald tests for linear combinations. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix H: Alternative-Explanation Models

Table H.1: Alternative-Explanation Models for the GDP–NTL Residual Profile

Model	Dependent variable: GDP–NTL residual profile		
	(1)	(2)	(3)
Sample	Non-connected / non-protégé	Hu and Xi eras	Xi era
Model	General Xi-era shift	Pooled decomposition	Pre-promotion NTL control
Xi era	0.096 (0.352) [0.787]	0.064 (0.336)	
Hu-connected PPS		0.254 (0.315)	
Additional Xi-era protégé gap		-1.318** (0.372) [0.002]	
Xi-protégé PPS			-1.040** (0.287) [0.015]
Pre-promotion NTL growth			0.033* (0.015)
Linear promotion-month trend	0.005 (0.004)	0.006 (0.004)	
Province fixed effects	Yes	Yes	Yes
Year fixed effects	No	No	Yes
Role fixed effects	Yes	Yes	Yes
Clustered standard errors	Province	Province	Province
Observations	1,477	1,916	657
Unique leaders	1,175	1,382	563
Provinces	25	25	25
Cities	268	271	253
Mayor promotions	791	1,023	332
Prefectural party-secretary promotions	686	893	325
Promotions by Hu-connected PPS	—	348	—
Promotions by non-connected PPS	910	910	—
Promotions by Xi-protégé PPS	—	91	91
Promotions by non-protégé PPS	567	567	566

Notes: The table reports alternative-explanation models for the GDP–NTL residual profile. Models (1) and (2) replace year fixed effects with a centered linear promotion-month trend so that the Xi-era indicator can be estimated. Model (1) restricts the sample to non-connected promoters in the Hu era and non-protégé promoters in the Xi era, and estimates the general Xi-era shift. Model (2) decomposes the pooled comparison into the general Xi-era shift, the Hu-era connected-promoter gap, and the additional Xi-era protégé gap. The additional Xi-era protégé gap refers to the Connected/protégé PPS $\times$ Xi era interaction term. Model (3) estimates the Xi-era comparison after controlling for average nighttime-light growth in the three years before promotion and retains year fixed effects. Promoter-type counts use the three-month buffer rule; dashes indicate categories outside the Xi-only sample. Standard errors clustered by province are shown in parentheses. Wild cluster bootstrap p-values are shown in square brackets for the coefficient of interest in each specification. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix I: Province- and Protégé-Level Influence Diagnostics

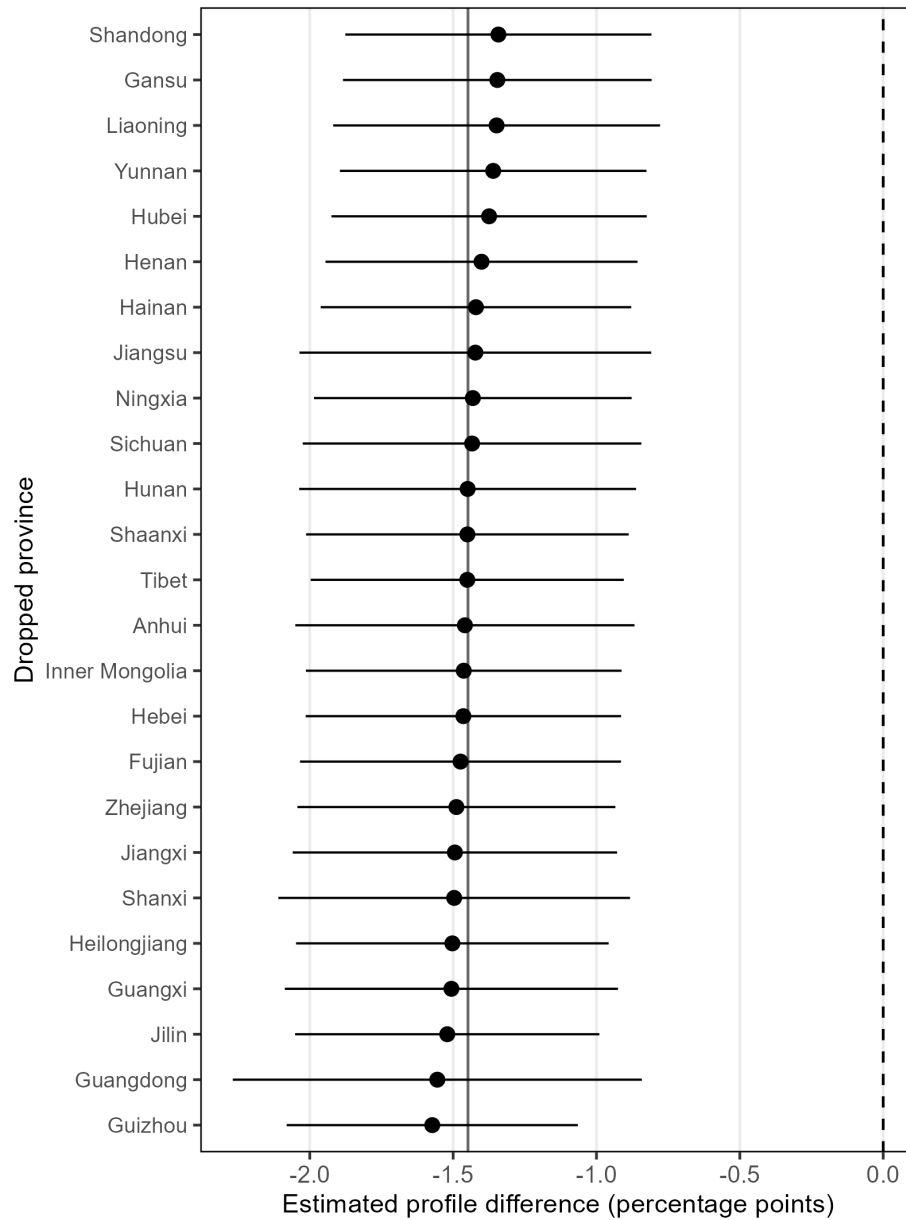


Figure I.1: Leave-One-Province-Out Estimates for the Pooled Hu–Xi Benchmark Model

Notes: Each point reports the Connected/protégé PPS $\times$ Xi era coefficient from the pooled Hu–Xi benchmark specification reported in Table F.3, column (4), re-estimated after dropping one province. Horizontal bars show 95% confidence intervals based on province-clustered standard errors. The solid vertical line marks the corresponding full-sample estimate; the dashed vertical line marks zero. More negative values indicate lower GDP–NTL residual profiles, meaning lower reported real GDP per capita growth relative to the nighttime-light benchmark.

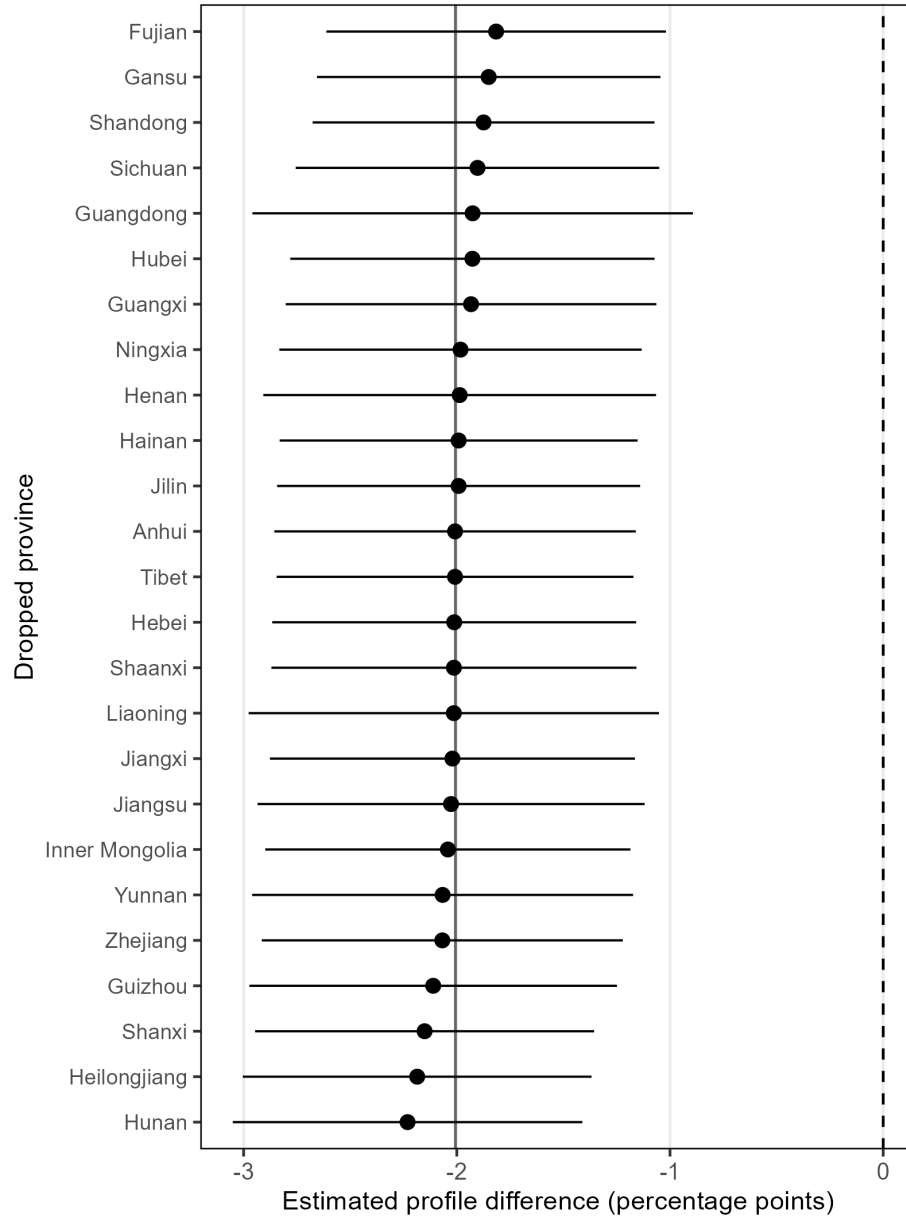


Figure I.2: Leave-One-Province-Out Estimates for Prefectural Party-Secretary Promotions

Notes: Each point reports the Connected/protégé PPS $\times$ Xi era coefficient from the specification for prefectural party-secretary promotions reported in Table G.1, column (1), re-estimated after dropping one province. Horizontal bars show 95% confidence intervals based on province-clustered standard errors. The solid vertical line marks the corresponding full-sample estimate; the dashed vertical line marks zero. More negative values indicate lower GDP–NTL residual profiles, meaning lower reported real GDP per capita growth relative to the nighttime-light benchmark.

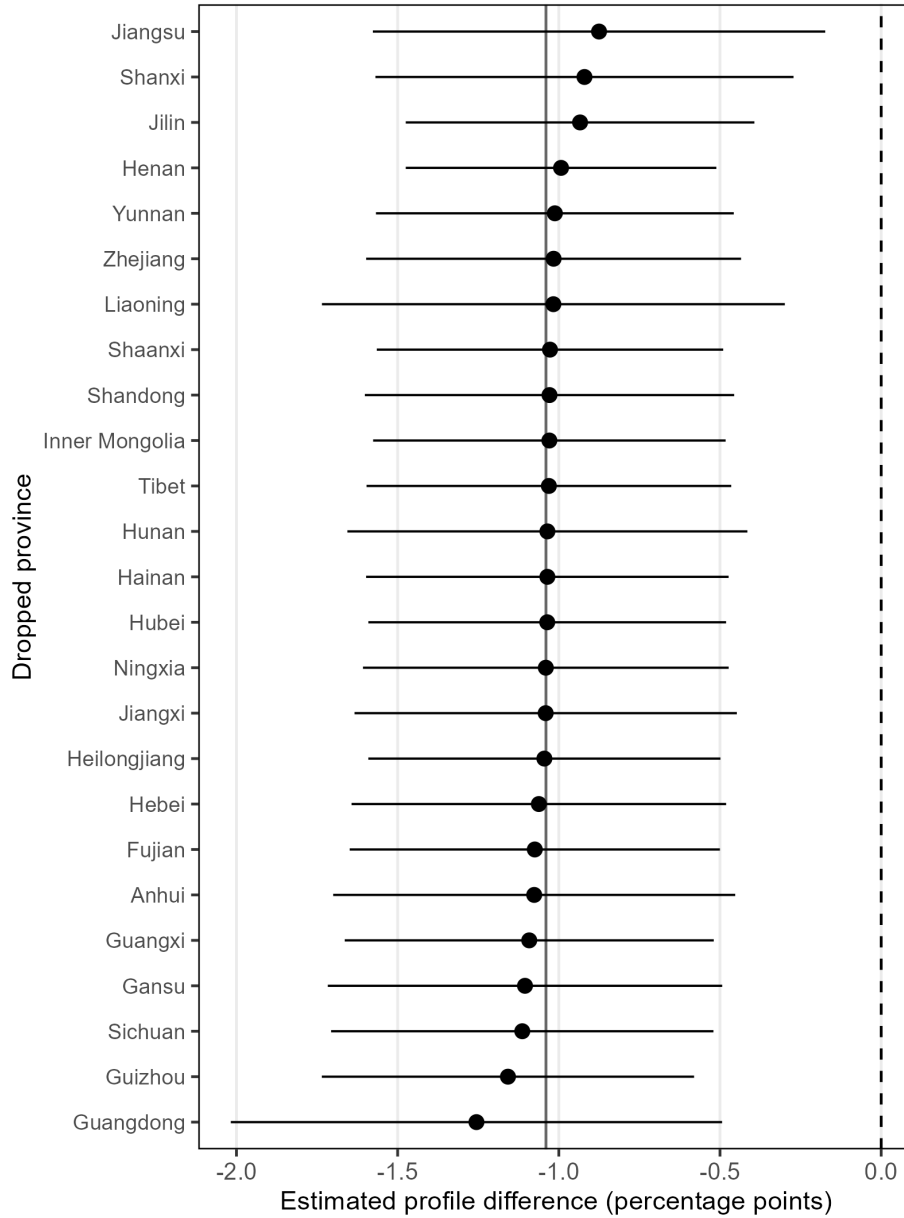


Figure I.3: Leave-One-Province-Out Estimates with Pre-Promotion NTL Growth Control

Notes: Each point reports the Xi-protégé PPS coefficient from the Xi-era pre-promotion NTL-control specification reported in Table H.1, column (3), re-estimated after dropping one province. Horizontal bars show 95% confidence intervals based on province-clustered standard errors. The solid vertical line marks the corresponding full-sample estimate; the dashed vertical line marks zero. More negative values indicate lower GDP–NTL residual profiles, meaning lower reported real GDP per capita growth relative to the nighttime-light benchmark.

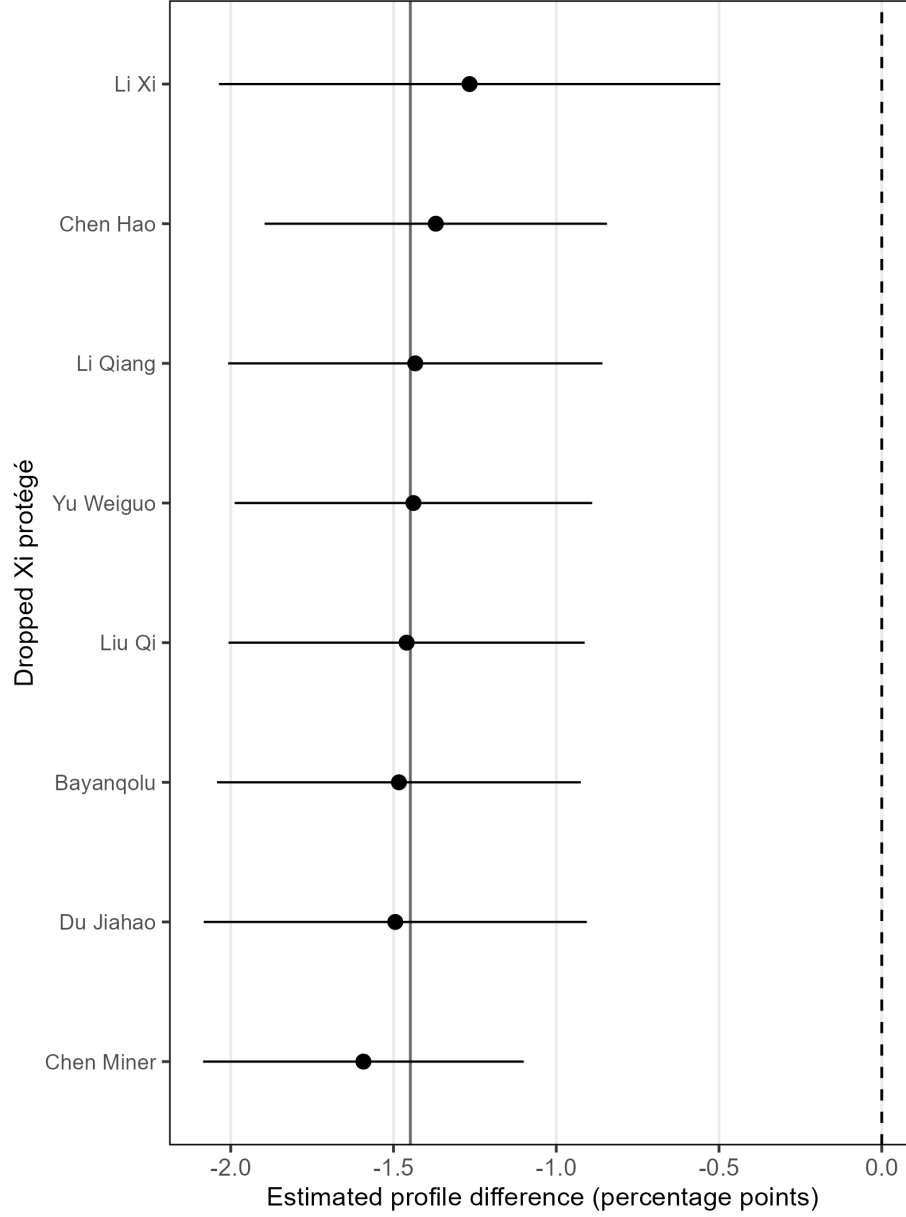


Figure I.4: Leave-One-Protégé-Out Estimates for the Pooled Hu–Xi Benchmark Model

Notes: Each point reports the Connected/protégé PPS $\times$ Xi era coefficient from the pooled Hu–Xi benchmark specification reported in Table F.3, column (4), re-estimated after dropping all Xi-era city-leader promotions attributed to one Xi-protégé provincial party secretary. Horizontal bars show 95% confidence intervals based on province-clustered standard errors. The solid vertical line marks the corresponding full-sample estimate; the dashed vertical line marks zero. More negative values indicate lower GDP–NTL residual profiles, meaning lower reported real GDP per capita growth relative to the nighttime-light benchmark.

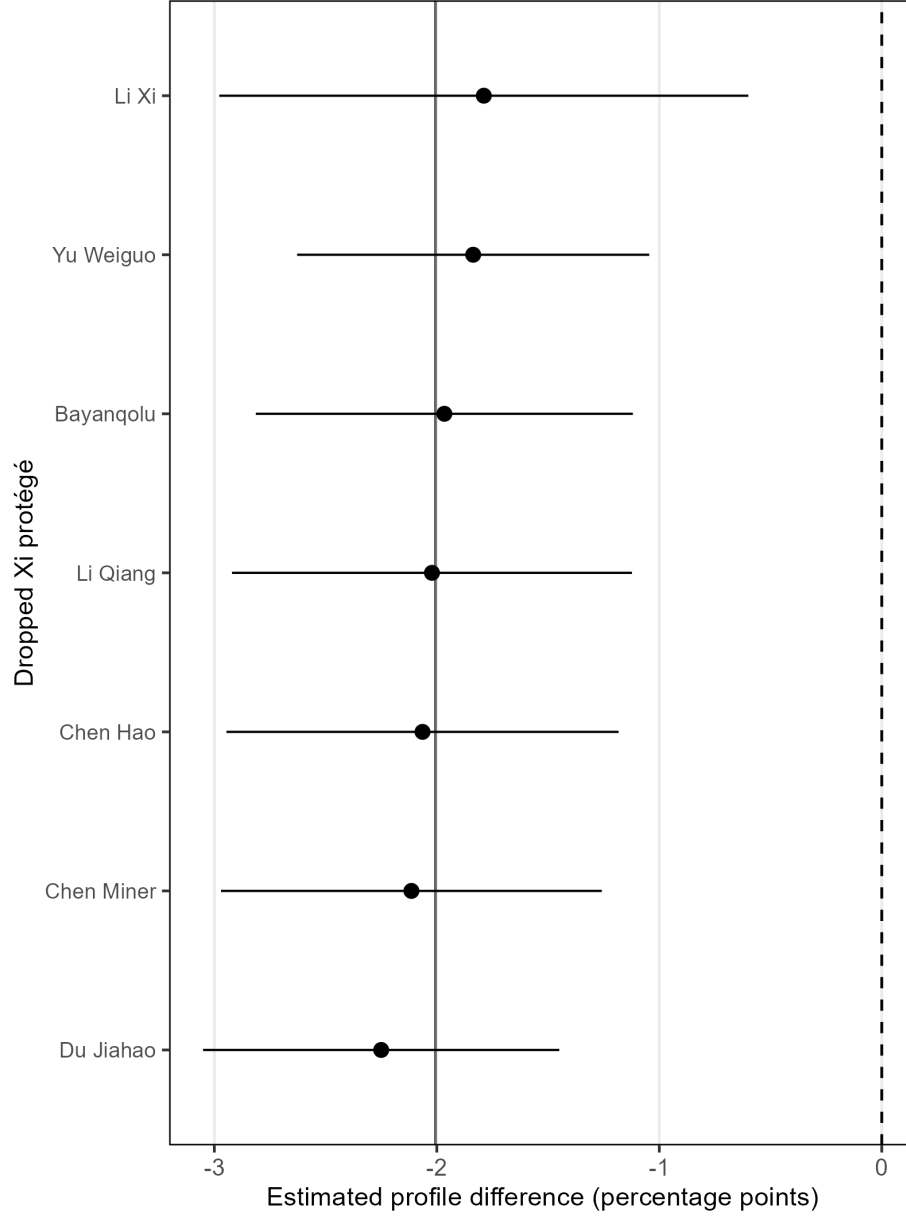


Figure I.5: Leave-One-Protégé-Out Estimates for Prefectural Party-Secretary Promotions

Notes: Each point reports the Connected/protégé PPS $\times$ Xi era coefficient from the specification for prefectural party-secretary promotions reported in Table G.1, column (1), re-estimated after dropping all Xi-era prefectural party-secretary promotions attributed to one Xi-protégé provincial party secretary. Horizontal bars show 95% confidence intervals based on province-clustered standard errors. The solid vertical line marks the corresponding full-sample estimate; the dashed vertical line marks zero. More negative values indicate lower GDP-NTL residual profiles, meaning lower reported real GDP per capita growth relative to the nighttime-light benchmark.

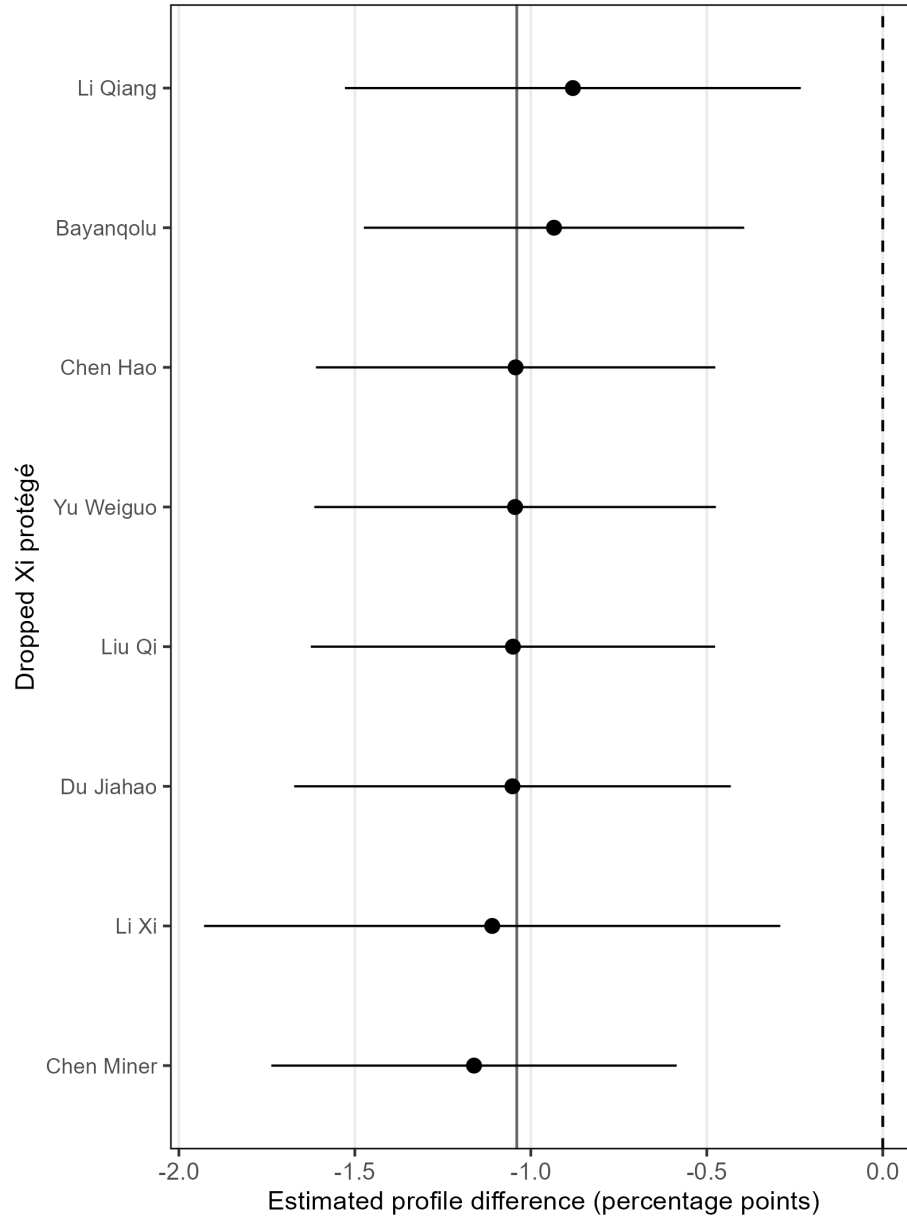


Figure I.6: Leave-One-Protégé-Out Estimates with Pre-Promotion NTL Growth Control

Notes: Each point reports the Xi-protégé PPS coefficient from the Xi-era pre-promotion NTL-control specification reported in Table H.1, column (3), re-estimated after dropping all city-leader promotions attributed to one Xi-protégé provincial party secretary. Horizontal bars show 95% confidence intervals based on province-clustered standard errors. The solid vertical line marks the corresponding full-sample estimate; the dashed vertical line marks zero. More negative values indicate lower GDP–NTL residual profiles, meaning lower reported real GDP per capita growth relative to the nighttime-light benchmark.

Appendix J: Electricity Robustness

Table J.1: First-Stage Residualization of Reported GDP per Capita Growth on Electricity Consumption Growth

Model	(1)
Dependent variable	Real GDP per capita growth
Electricity consumption growth	0.0004
(S.E.)	(0.0035)
City fixed effects	Yes
Year fixed effects	Yes
Clustered standard errors	City
City-year observations	4,067
Cities	282
Years	2005–2022
R-squared	0.432
Adjusted R-squared	0.387

Notes: The dependent variable is real GDP per capita growth. Real GDP per capita growth is trimmed at the 1st and 99th percentiles before residualization; the residual itself is not trimmed. The residual from this city-year model is used to estimate the leader-level GDP–electricity residual profile in Table J.2. Standard errors are clustered by city.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table J.2: Fixed-Effect Models Used to Estimate Electricity-Based Leader Profiles

Model	(1)	(2)
Dependent variable	Electricity consumption growth	GDP–electricity residual
Generated leader profile	Electricity consumption growth profile	GDP–electricity residual profile
Initial log level	-0.121***	-0.013*
(S.E.)	(0.015)	(0.006)
Log population	0.115	-0.201**
(S.E.)	(0.115)	(0.066)
Leader fixed effects	Yes	Yes
City fixed effects	Yes	Yes
Province-year fixed effects	Yes	Yes
Clustered standard errors	City	City
Observations	8,112	7,012
Unique leaders	1,843	1,517
Cities	272	260
Years	2005–2022	2005–2022
R-squared	0.606	0.591
Adjusted R-squared	0.431	0.417

Notes: The dependent variable in each column is the city-year outcome used to estimate the corresponding electricity-based leader fixed-effect profile. Entries report coefficient estimates with city-clustered standard errors in parentheses below estimates. Observations are leader-year-role observations because each city-year outcome is linked separately to the mayor and the prefectural party secretary. Electricity consumption growth is trimmed at the 1st and 99th percentiles. For the GDP–electricity residual outcome, real GDP per capita growth is trimmed before residualization and the residual itself is not trimmed. The generated profile is the centered leader fixed effect multiplied by 100. As an in-sample diagnostic, each full model is compared with an otherwise identical restricted model that omits leader fixed effects. Leader fixed effects are not jointly significant for electricity consumption growth ($F = 0.72$, $p = 1.000$), but are jointly significant for the GDP–electricity residual outcome ($F = 1.47$, $p < 0.001$).

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table J.3: Selection Models Using Electricity-Based Profiles

Model	Dependent variable: Electricity consumption growth profile				Dependent variable: GDP–electricity residual profile			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sample	Xi era	Hu era	Pooled: 3-month buffer	Pooled: no buffer	Xi era	Hu era	Pooled: 3-month buffer	Pooled: no buffer
Xi-protégé PPS	2.422 (1.353) [0.109]				-1.050** (0.351) [0.030]			
Hu-connected PPS		-0.537 (1.382) [0.719]				0.116 (0.279) [0.695]		
Xi era			Absorbed	Absorbed			Absorbed	Absorbed
Connected/protégé PPS			-0.056 (1.261)	-0.080 (1.292)			0.391 (0.242)	0.481 (0.266)
Connected/protégé PPS × Xi era			4.071 (2.211) [0.084]	4.658 (2.394) [0.057]			-1.303** (0.355) [0.004]	-1.375** (0.434) [0.022]
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Role fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered standard errors	Province	Province	Province	Province	Province	Province	Province	Province
Observations	711	1,216	1,927	1,927	592	1,156	1,748	1,748
Unique leaders	611	926	1,392	1,392	503	877	1,246	1,246
Provinces	22	22	22	22	22	21	22	22
Cities	246	258	259	259	226	242	245	245
Mayor promotions	371	671	1,042	1,042	296	637	933	933
Prefectural party-secretary promotions	340	545	885	885	296	519	815	815
Promotions by Hu-connected PPS	—	343	343	331	—	336	336	324
Promotions by non-connected PPS	—	873	873	885	—	820	820	832
Promotions by Xi-protégé PPS	111	—	111	121	79	—	79	83
Promotions by non-protégé PPS	600	—	600	590	513	—	513	509

Notes: The table reports robustness selection models using the two electricity-based leader profiles estimated in Table J.2. Columns (1)–(4) use the electricity consumption growth profile. Columns (5)–(8) use the GDP–electricity residual profile. Standard errors clustered by province are shown in parentheses. Wild cluster bootstrap p-values are shown in square brackets for the coefficient of interest in each specification. Columns (1)–(3) and (5)–(7) apply the three-month buffer rule; columns (4) and (8) use the no-buffer coding. Promoter-type counts follow the attribution rule used in each column, and dashes indicate categories outside an era-specific sample. All models include province, year, and role fixed effects; the Xi-era indicator is absorbed by year fixed effects in pooled specifications. Profiles are centered leader fixed effects multiplied by 100, so units are approximately percentage points. PPS denotes provincial party secretary.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix K: Stacked Event-Study Evidence from Prior Mayoral Spells

Table K.1: Stacked Event-Study Estimates for Fiscal Expenditure Growth

Event Time	Dependent variable: Fiscal expenditure growth			
	Window [-3,+3] Estimate (S.E.)	Later-selected event-years	Window [-4,+4] Estimate (S.E.)	Later-selected event-years
-4			-0.003 (0.015)	66
-3	0.009 (0.016)	66	0.011 (0.016)	66
-2	0.000 (0.018)	66	0.001 (0.018)	66
0	-0.033* (0.015)	67	-0.033* (0.015)	67
1	-0.023 (0.015)	67	-0.023 (0.015)	67
2	-0.015 (0.014)	67	-0.014 (0.014)	67
3	-0.016 (0.015)	67	-0.015 (0.015)	67
4			-0.013 (0.015)	58
Post-entry diff. (S.E.)	-0.025** (0.008)		-0.022** (0.008)	
Pre-trend test p	0.782		0.812	
Event Window	[-3,+3]		[-4,+4]	
Observations	2,713		3,428	
Events	391		392	
Later-selected Events	67		67	
Comparison Events	324		325	
Cities	212		213	
Calendar Years	2003–2024		2003–2024	
Adj. R2	0.387		0.395	
Within R2	0.004		0.003	
Event FE	Yes		Yes	
Year FE	Yes		Yes	
Event-time FE	Yes		Yes	

Notes: The table reports stacked event-study estimates for fiscal-expenditure growth during officials' most recent prior mayoral spells. Fiscal-expenditure growth is measured as the annual log change in city-level fiscal expenditure. The sample consists of officials who subsequently became first-time prefectural party secretaries between 2014 and 2024. The later-selected indicator equals one when the subsequent promotion is attributed to a Xi-protégé provincial party secretary under the three-month buffer rule; it classifies officials by subsequent selector type rather than treatment during the prior mayoral spell. The later-selected event-year columns count nonmissing later-selected observations at each event time; event totals vary with outcome- and window-specific sample restrictions. Reported event-time estimates are coefficients on interactions between the later-selected indicator and event-time indicators, with event time -1 omitted. The post-entry difference is estimated in a separate specification that replaces these interactions with a single later-selected $\times$ post-entry interaction. The pre-trend p -value tests the joint significance of the included pre-entry interactions. Standard errors are clustered by the city of the prior mayoral spell. All models include event, calendar-year, and event-time fixed effects.

· $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table K.2: Stacked Event-Study Estimates for Land-Revenue Expansion

Event Time	Dependent variable: Land-revenue expansion			
	Window [-3,+3] Estimate (S.E.)	Later-selected event-years	Window [-4,+4] Estimate (S.E.)	Later-selected event-years
-4			-0.133 (0.081)	69
-3	-0.071 (0.079)	69	-0.074 (0.078)	69
-2	-0.070 (0.079)	69	-0.074 (0.079)	69
0	-0.185* (0.083)	70	-0.185* (0.083)	70
1	-0.022 (0.078)	70	-0.025 (0.078)	70
2	-0.214** (0.083)	70	-0.214** (0.082)	70
3	-0.070 (0.076)	70	-0.077 (0.076)	70
4			-0.195* (0.087)	61
Post-entry diff. (S.E.)	-0.076 (0.040)		-0.068* (0.032)	
Pre-trend test p	0.565		0.376	
Event Window	[-3,+3]		[-4,+4]	
Observations	2,778		3,502	
Events	401		402	
Later-selected Events	70		70	
Comparison Events	331		332	
Cities	219		220	
Calendar Years	2003–2024		2003–2024	
Mean Outcome	0.571		0.567	
Adj. R2	0.154		0.157	
Within R2	0.004		0.004	
Event FE	Yes		Yes	
Year FE	Yes		Yes	
Event-time FE	Yes		Yes	

Notes: The table reports stacked event-study estimates for land-revenue expansion during officials' most recent prior mayoral spells. Land-revenue expansion equals one when observed city-level land-transfer revenue exceeds its value in the immediately preceding calendar year and zero otherwise; it is coded as missing when revenue is unavailable in either year. The sample consists of officials who subsequently became first-time prefectural party secretaries between 2014 and 2024. The later-selected indicator equals one when the subsequent promotion is attributed to a Xi-protégé provincial party secretary under the three-month buffer rule; it classifies officials by subsequent selector type rather than treatment during the prior mayoral spell. The later-selected event-year columns count nonmissing later-selected observations at each event time; event totals vary with outcome- and window-specific sample restrictions. Reported event-time estimates are coefficients on later-selected $\times$ event-time interactions, with event time -1 omitted. The post-entry difference comes from a separate specification using a single later-selected $\times$ post-entry interaction. The pre-trend p -value tests the joint significance of the included pre-entry interactions. Standard errors are clustered by the city of the prior mayoral spell. Estimates are reported in probability units. All models include event, calendar-year, and event-time fixed effects.

· $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.